ENGLISH FOR WORK / FAB / METROLOGY / QUALIFICATION / FOUNDRY

Semiconductor English

Name the layer. Qualify the result. Align the handoff.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 Wafer Fabrication Flow and Process Integration, page 4

Explain the current process stage and ownership without confusing wafer interconnect fabrication with package assembly.

02 Lithography, Reticles, and Critical Dimensions, page 16

Clarify a reported dimension change by identifying the layer, process stage, sampling, and measurement recipe before attributing a cause.

03 Deposition, Etch, CMP, and Process Windows, page 30

Describe a process trial improvement without extending it to untested conditions or overlooking downstream integration effects.

04 Metrology, SPC, and Yield Learning, page 44

Report yield with counts and comparison limits, and distinguish product acceptance from evidence about process stability.

05 Defect Density and Cleanroom Contamination, page 58

Describe inspection findings with location, area, timing, and classification while avoiding premature contamination-source claims.

06 Equipment Uptime, Recipes, and Tool Matching, page 70

Distinguish recipe identity, specification performance, tool matching, and equipment availability in a transfer discussion.

07 Packaging, Test, and Reliability Qualification, page 84

Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.

08 Foundry, Tape-Out, PDK, and Capacity Communication, page 95

Negotiate a requested date by separating design sign-off, data delivery, foundry acceptance, and capacity confirmation.

Professional communication notes, page 3

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Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

A number needs its process context.

Semiconductor discussions compress complex evidence into phrases such as CD shift, yield gain, matched tool, and tape-out ready. Useful communication restores the missing context: layer, stage, lot, measurement recipe, test coverage, or decision status. These cases practice precise technical questions and concise updates that help different specialties work from the same facts.

Locate the observation

Identify the layer, process stage, lot or wafer, tool, and time relevant to the claim. A change after etch is not automatically the same as a change measured in developed resist.
"The reported shift is on M2 after etch; the measurement recipe also changed."

Keep the denominator visible

Yield and defect figures depend on what was counted, sampled, and compared. State the population and measurement basis before interpreting the difference.
"The pilot result is 92 passing die out of 100 tested, with a different product mix."

Separate settings from performance

Identical recipe names do not prove tools behave alike. Compare the evidence against a defined matching criterion and distinguish availability from qualification.
"Both means are within specification, but the pair misses our matching limit."

Do not compress pending into complete

A finished test, a submitted layout, and a possible capacity slot each have a limited meaning. Name the remaining checks and who can authorize the next step.
"One qualification test passed; two required tests remain open."

Professional boundaries

Original fictional English practice, not a fabrication recipe, equipment procedure, product qualification, design sign-off, or production authorization. Numbers, lots, organizations, and schedules are invented. Real work must follow approved process documents, safety rules, applicable standards, customer requirements, and authorized engineering decisions.

A Read the situation

LESSON 01 / WAFER FABRICATION FLOW AND PROCESS INTEGRATION

Explain the current process stage and ownership without confusing wafer interconnect fabrication with package assembly.

> **Back-end does not always mean packaging**
>
> New product coordinator Eric describes lot L24 as being in packaging because the status says BEOL. Integration engineer Min explains that this lot is still in wafer fabrication, building on-wafer interconnect layers. Its current operation concerns metal layer M2. In this fictional route, an inspection review must be completed before the next authorized operation. Packaging has not started, and the lot is on hold pending that review. Eric must correct the customer update without inventing a shipping date.

People: Eric, Product coordinator; Min, Process integration engineer.

Read the record

Choose one answer per question using the facts above.

1. Where is L24 now?

A. In completed package assembly
B. In final shipment inspection
C. In customer field qualification
D. In wafer fabrication at an M2-related operation

2. What does BEOL mean in this case?

A. Back end of line, forming on-wafer interconnects
B. Completed assembly and final test
C. A confirmed customer delivery date
D. A shipping release status

3. What prevents the next operation?

A. A customer order cancellation
B. A recorded shipment approval
C. A pending inspection review and lot hold
D. A completed packaging qualification

Answers and explanations, page 124

B Vocabulary in use

CORE TERMS

wafer
A semiconductor substrate on which many devices are fabricated.

process a wafer

die
An individual circuit area or chip separated from a wafer.

identify the die

fab
A semiconductor fabrication facility.

move material through the fab

wafer fabrication
Processes that form devices and interconnects on wafers.

track wafer fabrication

process flow
The defined sequence of manufacturing operations.

review the process flow

process integration
Coordination of process steps into a functioning device flow.

assess process integration

FEOL
Front end of line, principally the formation of active devices.

complete FEOL processing

MOL
Middle of line, connecting devices to the interconnect structure.

review MOL interfaces

BEOL
Back end of line, forming the on-wafer interconnect structure.

track BEOL progress

interconnect
A conductive structure connecting circuit elements.

form an interconnect

metal layer
A defined level of conductive routing in a chip.

identify the metal layer

via
A conductive connection between interconnect levels.

inspect a via

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

wafer WAFER

process a wafer

Keep the meaning precise: A semiconductor substrate on which many devices are fabricated.

process integration PROCESS INTEGRATION

assess process integration

Keep the meaning precise: Coordination of process steps into a functioning device flow.

B Precision and professional shorthand

EXTEND YOUR RANGE

dielectric

An electrically insulating material used between conductors.

deposit a dielectric

lot

A tracked group of wafers or units processed under defined identity.

identify the lot

route

The controlled sequence assigned to a tracked product or lot.

follow the approved route

WIP

Work in process: material not yet through the defined manufacturing flow.

report WIP status

disposition

An authorized decision about the next treatment or status of material.

obtain lot disposition

dicing

Separating individual die from a wafer.

complete wafer dicing

process module

A related group of operations within the fabrication flow.

qualify a process module

wafer ID

The identifier assigned to an individual wafer.

confirm the wafer ID

traveler

A record accompanying material through its required operations.

update the traveler

lot hold

A restriction preventing a lot from proceeding without disposition.

place a lot on hold

wafer sort

Electrical testing of individual die while on the wafer.

schedule wafer sort

package assembly

Integrating and protecting die in the intended package structure.

begin package assembly

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

via VIA

inspect a via

Keep the meaning precise: A conductive connection between interconnect levels.

route ROUTE

follow the approved route

Keep the meaning precise: The controlled sequence assigned to a tracked product or lot.

C Phrases that move the work forward

Clarify the abbreviation
Here BEOL means back end of line.

Locate the work
The lot is still in wafer fabrication.

Name the layer
The current operation concerns metal layer M2.

Separate the stages
On-wafer interconnect formation is not package assembly.

Explain integration
We need to preserve compatibility between successive process modules.

Identify the material
Which lot and wafer IDs does the update cover?

Check the route
What is the next operation on the approved route?

State the hold
L24 is on hold pending inspection review.

Keep authority explicit
The next move needs authorized disposition.

Avoid a shipping inference
This stage does not establish a shipment date.

Distinguish the tests
Wafer sort and final packaged-device test are different steps.

Confirm the handoff
Which team owns the inspection review?

Correct the update
Packaging has not started for this lot.

Preserve traceability
Keep the traveler linked to the lot and operation.

Ask for the status basis
Does completed mean processed, inspected, or released?

Close precisely
Report the stage, hold reason, owner, and next decision.

Precision in real exchanges

Purpose: Separate the stages
Useful wording: On-wafer interconnect formation is not package assembly.

Purpose: Explain integration
Useful wording: We need to preserve compatibility between successive process modules.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which customer update is accurate?

A. L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.

B. L24 has completed the interconnect stack and awaits wafer sort.

C. L24 is in package assembly with final electrical test pending.

D. L24 is at M2 and can move while the inspection report is being completed.

2. Why expand BEOL for the coordinator?

A. BEOL always means package assembly in every organization.

B. Acronyms eliminate the need to name an operation.

C. The lot ID alone reveals its full status.

D. Different teams can use back-end for different parts of the manufacturing chain.

3. A colleague needs the term for this meaning: Processes that form devices and interconnects on wafers. Which term fits?

A. disposition

B. lot

C. BEOL

D. wafer fabrication

D Choose the response that fits / continued

4. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I will name M2 in the update. Does that identify the current layer, rather than mean the full interconnect stack is finished?" Min: [missing response] Eric: "Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab." Which response belongs in the gap?

A. FEOL principally forms the active devices, while later interconnect work connects them. The exact flow is technology-specific, so use our product route for the actual sequence and status.

B. Use wafer fabrication in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.

C. Here BEOL means back end of line. We are building the interconnect structure on the wafer, so your update places the lot at a later stage than the record supports.

D. Correct. M2 is the relevant metal layer. A layer name locates the work, but it does not by itself tell you that inspection, disposition, or later stages have finished.

Answers and explanations, page 124

One useful distinction

BEOL is still part of wafer fabrication in this conversation. It is not interchangeable with package assembly, although some organizations also use back-end for assembly and test. Expand the term and state the actual operation before communicating status.

E Conversation

Back-end does not always mean packaging

Back-end does not always mean packaging

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

BEOL / lot hold / process flow / disposition / package assembly / wafer fabrication / traveler / FEOL / metal layer / Wafer sort

01 **Eric:** I wrote that L24 is in packaging because the status says BEOL. The customer wants to know whether that means finished parts will be ready soon.

02 **Min:** Here [1] _____ means back end of line. We are building the interconnect structure on the wafer, so your update places the lot at a later stage than the record supports.

03 **Eric:** That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage.

04 **Min:** Use [2] _____ in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.

05 **Eric:** I will name M2 in the update. Does that identify the current layer, rather than mean the full interconnect stack is finished?

06 **Min:** Correct. M2 is the relevant [3] _____. A layer name locates the work, but it does not by itself tell you that inspection, disposition, or later stages have finished.

E Conversation / continued

07 **Eric:** Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab.

08 **Min:** [4] _____ principally forms the active devices, while later interconnect work connects them. The exact flow is technology-specific, so use our product route for the actual sequence and status.

09 **Eric:** I also see that inspection review is pending. The lot cannot simply move to its next listed operation because someone has already processed the current step.

10 **Min:** It is on a [5] _____. That restriction remains in place until the required review and authorized decision. Processed, inspected, and released are different status claims.

Conversation continues on the next page.

Back-end does not always mean packaging

CONVERSATION / CONTINUED

11 **Eric:** Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay.

12 **Min:** Yes. We need [6] _____ under the approved process. The review owner can explain the decision basis; we should not predict the outcome before that evidence is assessed.

13 **Eric:** I will confirm the owner and preserve the lot reference. We should also keep the record connected to the individual wafers where the inspection information requires that detail.

14 **Min:** Keep the [7] _____ and relevant wafer IDs aligned with the update. Otherwise two teams may discuss different material while assuming they are referring to the same issue.

15 **Eric:** What later stages should I distinguish in the message? I need to explain why an M2 status does not yet give the customer a finished-product date.

16 **Min:** Exactly. [8] _____, dicing, assembly, and subsequent checks have their own place in the approved route. Their timing is not established by the M2 entry alone.

17 **Eric:** I will correct the packaging statement now, before the customer plans around a stage the lot has not reached.

18 **Min:** Say that [9] _____ has not started for this lot. You can give a precise current status without inventing dates for the remaining operations.

19 **Eric:** The revised message will locate L24 in wafer fabrication at M2, explain the inspection hold, and identify the next review. It will leave shipment timing uncommitted.

20 **Min:** That reflects the [10] _____ accurately. The customer gets a clear explanation of progress and the pending decision instead of an optimistic interpretation of an abbreviation.

CHECK THE GAPS

1. BEOL / **2.** wafer fabrication / **3.** metal layer / **4.** FEOL / **5.** lot hold / **6.** disposition / **7.** traveler / **8.** Wafer sort / **9.** package assembly / **10.** process flow

Why these answers fit, page 125

F Check accuracy and scope

Back-end does not always mean packaging

Return to the case

New product coordinator Eric describes lot L24 as being in packaging because the status says BEOL. Integration engineer Min explains that this lot is still in wafer fabrication, building on-wafer interconnect layers. Its current operation concerns metal layer M2. In this fictional route, an inspection review must be completed before the next authorized operation. Packaging has not started, and the lot is on hold pending that review. Eric must correct the customer update without inventing a shipping date.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What is the appropriate next-step wording?

A. Promise shipment before checking the remaining flow.
B. The lot awaits inspection review and authorized disposition under its route.
C. Move the lot because M2 is usually near completion.
D. Skip the review because packaging has not started.

2. Which distinction is correct?

A. Wafer sort tests die on the wafer; package assembly integrates die into packages.
B. Wafer sort separates die mechanically; dicing classifies their electrical performance.
C. BEOL joins die to package substrates; package assembly forms on-wafer metal layers.
D. FEOL forms package connections; BEOL forms the active transistor devices.

Answers and explanations, page 127

Listen for the distinction

Switch roles. Read turns 9-20 again, stressing the hold, required decision, and stage not yet started.

Expand the abbreviation at a handoff

Back-end can refer to different stages in different teams. Use the full process name when crossing organizational boundaries, then identify the actual operation and lot status. Familiar shorthand is efficient only when the receiver shares its meaning.

F Say it / transfer to a new case

Name the stage before the date

Lot N6 is forming on-wafer interconnects. It is on hold for inspection review. Package assembly has not started, and no shipping date is confirmed.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

package assembly / BEOL / lot hold / disposition

01 **Coordinator:** I will correct the update: this on-wafer interconnect work is still [1] _____.

02 **Engineer:** Keep the restriction visible too; the record still shows a [2] _____.

03 **Coordinator:** Then I should not tell the customer we have started [3] _____.

04 **Engineer:** Correct. Ask the review owner for the authorized [4] _____ before discussing the next move.

CHECK THE GAPS

1. BEOL / **2.** lot hold / **3.** package assembly / **4.** disposition

Why these answers fit, page 127

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I will confirm the owner and preserve the lot reference. We should also keep the record connected to the individual wafers where the inspection information requires that detail." Min: [missing response] Eric: "What later stages should I distinguish in the message? I need to explain why an M2 status does not yet give the customer a finished-product date." Which response belongs in the gap?

A. Keep the traveler and relevant wafer IDs aligned with the update. Otherwise two teams may discuss different material while assuming they are referring to the same issue.

B. Use wafer fabrication in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.

C. FEOL principally forms the active devices, while later interconnect work connects them. The exact flow is technology-specific, so use our product route for the actual sequence and status.

D. Yes. We need disposition under the approved process. The review owner can explain the decision basis; we should not predict the outcome before that evidence is assessed.

Answers and explanations, page 128

A Read the situation

LESSON 02 / LITHOGRAPHY, RETICLES, AND CRITICAL DIMENSIONS

Clarify a reported dimension change by identifying the layer, process stage, sampling, and measurement recipe before attributing a cause.

A three-nanometer shift without a common basis

Lithography engineer Lina reviews a reported critical-dimension increase with metrology engineer Theo. The summary initially omits the layer and stage. The records show M2 after etch, with means of 86 nm and 89 nm. However, the two measurements used different CD-SEM recipes and different sampling sites. No comparable remeasurement, specification assessment, or cause investigation is complete. Lina must report the apparent 3 nm difference without treating it as a confirmed lithography shift or changing an exposure setting.

People: Lina, Lithography engineer; Theo, Metrology engineer.

Read the record

Choose one answer per question using the facts above.

1. Which layer and stage do the records identify?

A. Unspecified resist before exposure
B. Final package thickness
C. M2 before every fabrication step
D. M2 after etch

2. What is the reported numerical difference?

A. An increase of 89 nm
B. No numerical difference
C. An increase of 3 nm
D. A decrease of 3 nm

3. Why is a direct process attribution premature?

A. The layer is now identified.
B. Every post-etch result can only originate in lithography.
C. The measurement recipes and sampling sites differ.
D. Both figures use nanometers.

Answers and explanations, page 129

B Vocabulary in use

CORE TERMS

lithography
Pattern formation using exposure of a sensitive layer.

review lithography performance

photoresist
A radiation-sensitive material used to form a pattern.

coat the photoresist

focus offset
A displacement from a defined best-focus reference.

evaluate focus offset

CD-SEM
A scanning electron microscope used for critical-dimension measurement.

verify the CD-SEM recipe

sampling site
A specified location selected for measurement.

align sampling sites

post-etch
Measured or occurring after an etch operation.

compare post-etch measurements

reticle
A patterned optical element used in lithographic exposure.

identify the reticle

exposure dose
The delivered exposure energy per unit area.

record exposure dose

critical dimension
A feature dimension important to process or device performance, abbreviated CD.

measure critical dimension

measurement recipe
The controlled settings and analysis used to obtain a measurement.

confirm the measurement recipe

post-develop
Measured or occurring after resist development.

review post-develop CD

etch bias
A dimensional difference associated with pattern transfer through etch.

evaluate etch bias

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

lithography LITHOGRAPHY

review lithography performance

Keep the meaning precise: Pattern formation using exposure of a sensitive layer.

critical dimension CRITICAL DIMENSION

measure critical dimension

Keep the meaning precise: A feature dimension important to process or device performance, abbreviated CD.

B Precision and professional shorthand

EXTEND YOUR RANGE

overlay

Alignment accuracy between patterned layers.

measure overlay

alignment mark

A feature used to position a layer or measurement.

inspect alignment marks

line-edge roughness

Variation of a patterned line edge from its intended shape.

assess line-edge roughness

line-width roughness

Variation in the width of a patterned line.

assess line-width roughness

pitch

The repeat distance between corresponding features.

specify feature pitch

pattern fidelity

Agreement between the intended and produced pattern.

assess pattern fidelity

optical proximity correction

Layout adjustments compensating for imaging effects, abbreviated OPC.

review optical proximity correction

focus-exposure matrix

A test pattern set spanning selected focus and dose conditions.

analyze a focus-exposure matrix

depth of focus

A usable focus range under a defined imaging criterion.

assess depth of focus

exposure latitude

A usable dose range under a defined imaging criterion.

estimate exposure latitude

measurement bias

A systematic measurement difference relative to a reference.

assess measurement bias

remeasurement

A repeated measurement under identified conditions.

request comparable remeasurement

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

etch bias ETCH BIAS

evaluate etch bias

Keep the meaning precise: A dimensional difference associated with pattern transfer through etch.

pitch PITCH

specify feature pitch

Keep the meaning precise: The repeat distance between corresponding features.

C Phrases that move the work forward

Locate the dimension
Which layer and feature does the CD refer to?

Identify the stage
These measurements are from M2 after etch.

State the numbers
The reported means are 86 and 89 nm.

Qualify the difference
The apparent difference is 3 nm.

Ask about measurement
Did both datasets use the same CD-SEM recipe?

Check sampling
Were the same types of sites sampled?

Avoid a causal shortcut
Post-etch CD does not isolate exposure performance.

Separate metrics
Overlay is not another name for critical dimension.

Request comparability
Remeasure on an agreed, traceable basis.

Preserve the originals
Keep both original recipes with their datasets.

Avoid an unsupported adjustment
Do not infer an exposure change from this comparison alone.

Ask for limits
Which specification applies to this feature and stage?

Distinguish variation
A mean shift and within-wafer variation are different questions.

Clarify reticle identity
Which reticle revision was used for the layer?

Describe the next check
Resolve the measurement difference before attributing the process cause.

Close the report
State the apparent shift and the comparability limitation together.

Precision in real exchanges

Purpose: Qualify the difference
Useful wording: The apparent difference is 3 nm.

Purpose: Ask about measurement
Useful wording: Did both datasets use the same CD-SEM recipe?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which report sentence is justified?

A. Both datasets prove the same physical result because they use nanometers.

B. M2 is out of specification without any stated limit.

C. The means differ by 3 nm, but recipe and site changes limit comparability.

D. Exposure drift has been confirmed at exactly 3 nm.

2. Which comparison best helps investigate the apparent change?

A. Increasing sample counts while leaving recipe and site differences uncontrolled

B. Comparing the post-etch mean with a post-develop mean as though the stages were identical

C. Comparable measurements on identified sites using an agreed recipe basis

D. Repeating both datasets with their different original recipes and locations

3. A colleague needs the term for this meaning: The delivered exposure energy per unit area. Which term fits?

A. line-edge roughness

B. sampling site

C. depth of focus

D. exposure dose

D Choose the response that fits / continued

4. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting." Theo: [missing response] Lina: "I will keep the three-nanometer difference and remove the exposure-drift conclusion. Can we show the changed recipe and sites beside the two means?" Which response belongs in the gap?

A. The sampling site selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.

B. That protects pattern fidelity discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

C. Exactly. Etch bias is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

D. Request comparable remeasurement through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.

Answers and explanations, page 129

One useful distinction

The arithmetic difference is 3 nm. That does not prove a 3 nm physical process shift because the recipes and sampled sites differ. A post-etch CD combines the history of pattern formation and transfer; it does not identify exposure as the sole cause.

E Conversation

A three-nanometer shift without a common basis

A three-nanometer shift without a common basis

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

remeasurement / post-etch / sampling site / Overlay / Etch bias / critical dimension / measurement bias / reticle / measurement recipe / pattern fidelity

01 **Lina:** The review slide says CD shifted, but it does not identify the layer or the measurement stage. I cannot tell which process team should investigate first.

02 **Theo:** The records identify [1] _____ M2 measurements. The two means are eighty-six and eighty-nine nanometers, so the reported numerical increase is three nanometers.

03 **Lina:** That locates the result, but I still need to know whether the two measurements were obtained on a comparable basis before calling it a physical process shift.

04 **Theo:** The [2] _____ changed between datasets. Both used CD-SEM, but sharing the instrument type does not mean their settings and analysis were identical.

05 **Lina:** Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting.

06 **Theo:** The [3] _____ selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.

E Conversation / continued

07 **Lina:** I will keep the three-nanometer difference and remove the exposure-drift conclusion. Can we show the changed recipe and sites beside the two means?

08 **Theo:** Correct. The [4] _____ result is important, but the present comparison does not separate process behavior from measurement and sampling effects.

09 **Lina:** Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement.

10 **Theo:** Exactly. [5] _____ is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

Conversation continues on the next page.

A three-nanometer shift without a common basis

CONVERSATION / CONTINUED

11 **Lina:** The review also asks whether this is an alignment problem. Can we clarify that separately? I do not want the CD figure copied into an overlay conclusion.

12 **Theo:** [6] _____ concerns alignment between layers. It is not another name for feature width, so this CD comparison cannot be relabeled as an overlay finding.

13 **Lina:** Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis.

14 **Theo:** Request comparable [7] _____ through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.

15 **Lina:** If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy.

16 **Theo:** Yes. Assess [8] _____ and other relevant effects rather than force either outcome. The aim is a supported interpretation, not a preferred conclusion that the process changed or stayed identical.

17 **Lina:** I will include the reticle revision in the context record as well. That gives the investigation the configuration information without implying the reticle has been found defective.

18 **Theo:** Good. The [9] _____ identity belongs with the layer history. Configuration details help evaluate hypotheses, but listing a component does not make it the cause.

A three-nanometer shift without a common basis / continued

19 **Lina:** The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open.

20 **Theo:** That protects [10] _____ discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

CHECK THE GAPS

1. post-etch / **2.** measurement recipe / **3.** sampling site / **4.** critical dimension / **5.** Etch bias / **6.** Overlay / **7.** remeasurement / **8.** measurement bias / **9.** reticle / **10.** pattern fidelity

Why these answers fit, page 131

F Check accuracy and scope

A three-nanometer shift without a common basis

Return to the case

Lithography engineer Lina reviews a reported critical-dimension increase with metrology engineer Theo. The summary initially omits the layer and stage. The records show M2 after etch, with means of 86 nm and 89 nm. However, the two measurements used different CD-SEM recipes and different sampling sites. No comparable remeasurement, specification assessment, or cause investigation is complete. Lina must report the apparent 3 nm difference without treating it as a confirmed lithography shift or changing an exposure setting.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why is post-etch important?

A. It makes the sampling locations irrelevant.
B. The result follows pattern transfer and does not isolate lithography alone.
C. It means no etch process has occurred.
D. It establishes that every cause is in the reticle.

2. Which distinction is accurate?

A. CD concerns feature size; overlay concerns layer alignment.
B. CD measures layer alignment; overlay measures a line's width.
C. Pitch measures line-edge variation; line-edge roughness measures repeat distance.
D. Etch bias is the exposure energy; exposure dose is the post-etch dimensional change.

Answers and explanations, page 132

Listen for the distinction

Switch roles for turns 7-16. Stress apparent difference, post-etch, and the changed measurement basis.

Ask which dimension was measured

CD shift sounds specific but can conceal different features, stages, or measurement settings. A useful challenge asks for the measurement basis before debating process causes. Do not assume that a post-etch result isolates lithography or that a changed mean proves a specification violation.

F Say it / transfer to a new case

Same units, different measurement basis

A layer has reported means of 120 and 124 nm after etch. Different measurement recipes and sampling sites were used. The cause is not established.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

critical dimension / sampling sites / overlay / measurement recipe

01 **Engineer:** Check that both reports refer to the same [1] _____.

02 **Reviewer:** Send the settings and analysis from each [2] _____.

03 **Engineer:** I will attach a wafer map of both sets of [3] _____.

04 **Reviewer:** Good. An alignment claim would need separate [4] _____ evidence.

CHECK THE GAPS

1. critical dimension / **2.** measurement recipe / **3.** sampling sites / **4.** overlay

Why these answers fit, page 132

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis." Theo: [missing response] Lina: "If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy." Which response belongs in the gap?

A. Exactly. Etch bias is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

B. The sampling site selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.

C. That protects pattern fidelity discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

D. Request comparable remeasurement through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.

Answers and explanations, page 133

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Wafer Fabrication Flow and Process Integration" and "Lithography, Reticles, and Critical Dimensions". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A semiconductor fabrication facility. Which term fits?

A. fab

B. process module

C. MOL

D. lot hold

2. A colleague needs the term for this meaning: The controlled settings and analysis used to obtain a measurement. Which term fits?

A. lithography

B. optical proximity correction

C. overlay

D. measurement recipe

3. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I also see that inspection review is pending. The lot cannot simply move to its next listed operation because someone has already processed the current step." Min: [missing response] Eric: "Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay." Which response belongs in the gap?

A. It is on a lot hold. That restriction remains in place until the required review and authorized decision. Processed, inspected, and released are different status claims.

B. Use wafer fabrication in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.

C. That reflects the process flow accurately. The customer gets a clear explanation of progress and the pending decision instead of an optimistic interpretation of an abbreviation.

D. Exactly. Wafer sort, dicing, assembly, and subsequent checks have their own place in the approved route. Their timing is not established by the M2 entry alone.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "I will include the reticle revision in the context record as well. That gives the investigation the configuration information without implying the reticle has been found defective." Theo: [missing response] Lina: "The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open." Which response belongs in the gap?

A. Good. The reticle identity belongs with the layer history. Configuration details help evaluate hypotheses, but listing a component does not make it the cause.

B. Overlay concerns alignment between layers. It is not another name for feature width, so this CD comparison cannot be relabeled as an overlay finding.

C. That protects pattern fidelity discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

D. The measurement recipe changed between datasets. Both used CD-SEM, but sharing the instrument type does not mean their settings and analysis were identical.

Answers and explanations, page 164

A Read the situation

LESSON 03 / DEPOSITION, ETCH, CMP, AND PROCESS WINDOWS

Describe a process trial improvement without extending it to untested conditions or overlooking downstream integration effects.

> **A better center point is not a wider window**
>
> Deposition engineer Yuki and integration engineer Sam review a trial film process. At the nominal setting, a defined thickness-nonuniformity metric falls from 5% to 3%; lower is better on this unchanged measurement basis. Only that center condition has been tested. The proposed low and high settings remain untested, and effects on subsequent etch and chemical mechanical planarization are not assessed. The new recipe is not released. A summary nevertheless says the complete process window is validated.

People: Yuki, Deposition engineer; Sam, Integration engineer.

Read the record

Choose one answer per question using the facts above.

1. What improved in the supplied trial?

A. The new recipe received production release.
B. The stated nonuniformity metric fell from 5% to 3% at the nominal setting.
C. Every condition in the proposed range was tested.
D. Etch and planarization were fully qualified.

2. What is the absolute change in that percentage metric?

A. A decrease of 2 percentage points
B. A decrease of 2 relative percent
C. An increase of 3 percentage points
D. A decrease of 5 percentage points

3. Which conclusion is premature?

A. The low and high settings remain untested.
B. Downstream integration effects remain open.
C. The complete process window is validated.
D. The nominal-setting result improved on the stated metric.

Answers and explanations, page 134

B Vocabulary in use

CORE TERMS

deposition
Formation of a material film on a surface.

qualify a deposition step

thin film
A material layer with a small controlled thickness.

characterize the thin film

CVD
Chemical vapor deposition, forming a film through chemical reactions of vapor precursors.

evaluate CVD film properties

PVD
Physical vapor deposition, forming a film from material vaporized by methods such as sputtering or evaporation.

review PVD performance

ALD
Atomic layer deposition, using successive self-limiting surface reactions.

assess ALD conformality

precursor
A starting substance used in a film-forming reaction.

qualify the precursor

conformality
How consistently a film coats different surface orientations or features.

assess film conformality

step coverage
Film thickness on a specified feature relative to thickness on a reference surface.

evaluate step coverage

film stress
Internal stress within a deposited film.

measure film stress

thickness nonuniformity
Variation in film thickness under a defined metric.

report thickness nonuniformity

etch selectivity
Relative removal rates of different materials in an etch process.

assess etch selectivity

anisotropy
Direction-dependent behavior, such as preferential vertical removal.

evaluate etch anisotropy

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

deposition DEPOSITION

qualify a deposition step

Keep the meaning precise: Formation of a material film on a surface.

precursor PRECURSOR

qualify the precursor

Keep the meaning precise: A starting substance used in a film-forming reaction.

B Precision and professional shorthand

EXTEND YOUR RANGE

endpoint detection

Monitoring used to identify a process completion condition.

verify endpoint detection

CMP

Chemical mechanical planarization, combining chemical and mechanical surface removal.

evaluate CMP compatibility

dishing

Local depression of a material surface during planarization.

measure dishing

process window

A range of conditions meeting defined process requirements.

establish the process window

corner condition

A selected limiting combination in an evaluation range.

evaluate corner conditions

overetch

Additional etching beyond a defined clearing or endpoint condition.

assess overetch effects

slurry

A polishing fluid mixture supplying chemical action and often suspended abrasive particles in CMP.

qualify the CMP slurry

erosion

Excess regional material removal during planarization.

assess erosion

nominal setting

The chosen central or reference process setting.

test the nominal setting

process interaction

An effect arising from the relationship between process steps or factors.

investigate process interactions

B Precision and professional shorthand / continued

integration trade-off

An exchange between benefits and effects across process steps.

explain the integration trade-off

recipe release

Authorization of a controlled process recipe for defined use.

obtain recipe release

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

anisotropy ANISOTROPY

evaluate etch anisotropy

Keep the meaning precise: Direction-dependent behavior, such as preferential vertical removal.

dishing DISHING

measure dishing

Keep the meaning precise: Local depression of a material surface during planarization.

C Phrases that move the work forward

Identify the metric

The defined thickness-nonuniformity metric fell from 5% to 3%.

Keep the basis fixed

Both values use the same measurement definition.

State the condition

The result applies to the nominal setting only.

Separate changes

That is two percentage points, or a 40% relative reduction.

Limit the claim

One center point does not establish a process window.

Name the missing tests

The proposed low and high settings are untested.

Check interactions

What does the film change mean for the subsequent etch?

Ask about planarization

Has compatibility with the CMP step been assessed?

Avoid a causal leap

Better uniformity does not prove every film property improved.

Identify trade-offs

We still need the integration trade-off assessment.

Qualify maturity

This is a trial result, not a released recipe.

Ask for criteria

Which responses must meet limits across the range?

Preserve configuration

Keep the trial recipe revision with the measurement record.

Request bounded follow-up

Assess the planned range and the identified downstream effects.

Avoid declaring failure

Untested corners are unresolved, not failed.

Close the summary

Report the center-point result and the remaining validation scope.

Precision in real exchanges

Purpose: Separate changes
Useful wording: That is two percentage points, or a 40% relative reduction.

Purpose: Limit the claim
Useful wording: One center point does not establish a process window.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary preserves the evidence boundary?

A. The trial establishes the window, with etch and CMP review needed only after release.

B. The nominal-setting metric improved; the range and downstream effects remain untested.

C. The metric improved at nominal, so the same reduction applies at the low and high settings.

D. A lower nonuniformity metric establishes improved film stress and conformality too.

2. What is the relative reduction from 5% to 3%?

A. 60%

B. 3%

C. 40%

D. 2%

3. A colleague needs the term for this meaning: Physical vapor deposition, forming a film from material vaporized by methods such as sputtering or evaporation. Which term fits?

A. PVD

B. CMP

C. corner condition

D. film stress

D Choose the response that fits / continued

4. Rebuild the exchange from "A better center point is not a wider window". Yuki: "The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent." Sam: [missing response] Yuki: "I will mark the low and high settings untested. What evidence does the approved plan require before we can make a claim about that range?" Which response belongs in the gap?

A. Name thickness nonuniformity explicitly. Saying the process improved could imply that every relevant response was assessed, while the reported result addresses only this defined metric.

B. That presents the deposition result accurately. The team can recognize progress while seeing exactly what evidence is still needed before making a broader process claim.

C. Keep that arithmetic tied to the nominal setting. Neither way of expressing the improvement extends the evidence to settings that have not been run.

D. Keep recipe release separate from the experiment. A favorable metric can support the review, but the required scope and approvals still need to be completed.

Answers and explanations, page 134

> **One useful distinction**
>
> The metric improves by 2 percentage points, a 40% relative reduction from its initial 5% value. Neither calculation establishes the proposed window. The same measurement definition is used here; other nonuniformity formulas must not be mixed into the comparison.

E Conversation

A better center point is not a wider window

A better center point is not a wider window

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

process window / etch selectivity / deposition / corner condition / thickness nonuniformity / recipe release / process interaction / integration trade-off / CMP / nominal setting

01 **Yuki:** The nominal-setting trial reduced our thickness-nonuniformity metric from five percent to three percent. The summary now says the full process window is validated, which seems too broad.

02 **Sam:** It is too broad. A [1] _____ covers a range of conditions meeting defined requirements. We have one encouraging center-point result, not evidence for the entire proposed range.

03 **Yuki:** Both values use the same measurement definition, and lower is better for this metric. I want to report that improvement clearly without losing the limitation.

04 **Sam:** Name [2] _____ explicitly. Saying the process improved could imply that every relevant response was assessed, while the reported result addresses only this defined metric.

05 **Yuki:** The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent.

06 **Sam:** Keep that arithmetic tied to the [3] _____. Neither way of expressing the improvement extends the evidence to settings that have not been run.

E Conversation / continued

07 **Yuki:** I will mark the low and high settings untested. What evidence does the approved plan require before we can make a claim about that range?

08 **Sam:** Correct. Each planned [4] _____ needs its own assessment under the approved plan. An untested condition is unresolved, not a favorable result or a demonstrated defect.

09 **Yuki:** The deposition change may also affect other film characteristics. A lower thickness-variation metric does not tell us that stress, coverage, or later pattern transfer remained acceptable.

10 **Sam:** That is an [5] _____ question. We need to assess the relevant responses and interactions rather than optimize one number in isolation from the product flow.

Conversation continues on the next page.

A better center point is not a wider window

CONVERSATION / CONTINUED

11 **Yuki:** For the subsequent etch, the team has not yet checked how the changed film behaves relative to the neighboring materials. That could affect the interpretation of the trial.

12 **Sam:** Review [6] _____ where it matters to the approved integration assessment. Naming the concern identifies a question to investigate, not a result we can claim already occurred.

13 **Yuki:** The planarization team also needs the proposed film information. Their review has not started, so I should not imply that the next surface-finishing step is already covered.

14 **Sam:** State that [7] _____ compatibility remains open. The deposition result does not establish that planarization behavior, surface effects, or related requirements are unchanged.

15 **Yuki:** We can identify the relevant responses in the evaluation plan without improvising operating settings during this review. The process owners should define the approved technical work.

16 **Sam:** Yes. Capture each [8] _____ question and its owner. That makes the remaining work specific enough to schedule and review without treating the meeting as recipe authorization.

17 **Yuki:** Please keep trial revision on the summary. Planning is asking about routine use, and I do not want this result mistaken for a production release.

18 **Sam:** Keep [9] _____ separate from the experiment. A favorable metric can support the review, but the required scope and approvals still need to be completed.

A better center point is not a wider window / continued

19 **Yuki:** I will revise the summary to report the center-point improvement, the unchanged metric basis, and the untested range. I will list the etch and planarization reviews as pending.

20 **Sam:** That presents the [10] _____ result accurately. The team can recognize progress while seeing exactly what evidence is still needed before making a broader process claim.

CHECK THE GAPS

1. process window / **2.** thickness nonuniformity / **3.** nominal setting / **4.** corner condition / **5.** integration trade-off / **6.** etch selectivity / **7.** CMP / **8.** process interaction / **9.** recipe release / **10.** deposition

Why these answers fit, page 136

F Check accuracy and scope

A better center point is not a wider window

Return to the case

Deposition engineer Yuki and integration engineer Sam review a trial film process. At the nominal setting, a defined thickness-nonuniformity metric falls from 5% to 3%; lower is better on this unchanged measurement basis. Only that center condition has been tested. The proposed low and high settings remain untested, and effects on subsequent etch and chemical mechanical planarization are not assessed. The new recipe is not released. A summary nevertheless says the complete process window is validated.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why review downstream etch and CMP?

A. Lower thickness variation removes the need to assess other film properties.
B. A film change may alter later process behavior.
C. The deposition metric already establishes the etch selectivity.
D. An unchanged CMP recipe proves the new film polishes identically.

2. Which status should remain separate from trial success?

A. The recorded original value of 5%
B. Recipe release
C. The reported nominal-setting value of 3%
D. The unchanged metric definition

Answers and explanations, page 137

Listen for the distinction

Switch roles. Read the etch and CMP exchanges while keeping both assessments pending.

Celebrate a bounded improvement

An encouraging center-point result is worth reporting. State the metric and tested condition, then identify the untested range and integration questions. This lets colleagues recognize progress without confusing a promising experiment with a released process window.

F Say it / transfer to a new case

An improved metric with a limited range

At one nominal setting, an unchanged nonuniformity metric falls from 8% to 6%. Lower is better. Edge settings and downstream etch remain untested; the recipe is not released.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

recipe release / integration question / nominal setting / process window

01 **Engineer:** The metric fell from eight to six percent, but only at the [1] _____.

02 **Reviewer:** Keep the low and high settings open; we have not established the [2] _____.

03 **Engineer:** I will also list the untested effect on the later etch as an [3] _____.

04 **Reviewer:** And keep routine production use pending the authorized [4] _____.

CHECK THE GAPS

1. nominal setting / **2.** process window / **3.** integration question / **4.** recipe release

Why these answers fit, page 137

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A better center point is not a wider window". Yuki: "The planarization team also needs the proposed film information. Their review has not started, so I should not imply that the next surface-finishing step is already covered." Sam: [missing response] Yuki: "We can identify the relevant responses in the evaluation plan without improvising operating settings during this review. The process owners should define the approved technical work." Which response belongs in the gap?

A. Keep that arithmetic tied to the nominal setting. Neither way of expressing the improvement extends the evidence to settings that have not been run.

B. Name thickness nonuniformity explicitly. Saying the process improved could imply that every relevant response was assessed, while the reported result addresses only this defined metric.

C. State that CMP compatibility remains open. The deposition result does not establish that planarization behavior, surface effects, or related requirements are unchanged.

D. Keep recipe release separate from the experiment. A favorable metric can support the review, but the required scope and approvals still need to be completed.

Answers and explanations, page 138

A Read the situation

LESSON 04 / METROLOGY, SPC, AND YIELD LEARNING

Report yield with counts and comparison limits, and distinguish product acceptance from evidence about process stability.

Two percentage points, not a proven process gain

Yield engineer Ravi presents a pilot lot with 92 passing die out of 100 tested, compared with a reference lot with 900 passing die out of 1,000 tested. The product mixes differ, and no controlled comparison or statistical significance assessment has been completed. Manager Elena wants to announce that a process change caused a two-percent yield improvement. The team also has no time-ordered control-chart analysis supporting a stability claim. Ravi must preserve the useful observations while correcting the causal, percentage, and stability language.

People: Elena, Yield program manager; Ravi, Yield engineer.

Read the record

Choose one answer per question using the facts above.

1. What is the pilot yield?

A. 100% because the pilot completed
B. 92% from 92 of 100 tested die
C. 90% from 900 of 1,000 die
D. 2% from two lots

2. What is the absolute difference between the observed yields?

A. Exactly 2 relative percent
B. 20 percentage points
C. 2 percentage points
D. 2 passing die

3. Which factor limits the causal comparison?

A. The pilot is called a pilot.
B. The reference contains more passing die in absolute number.
C. The product mixes differ.
D. Both yields have denominators.

Answers and explanations, page 139

B Vocabulary in use

CORE TERMS

metrology
The science and practice of measurement.

review metrology consistency

pass count
The number of tested items meeting the stated criteria.

preserve the pass count

denominator
The total used as the base of a fraction or percentage.

state the denominator

pilot lot
A limited production lot used to evaluate a process or product.

evaluate a pilot lot

percentage point
A unit for the absolute difference between percentages.

report percentage-point change

confounding factor
A factor that prevents clean attribution of an observed relationship.

identify a confounding factor

electrical yield
The proportion meeting defined electrical test criteria.

report electrical yield

tested population
The set of items included in a reported test result.

define the tested population

product mix
The composition of different products in a dataset.

adjust for product mix

reference lot
A lot used as a stated comparison basis.

identify the reference lot

relative change
A change divided by its specified starting value.

calculate relative change

stratification
Separating data into relevant comparable groups.

stratify by product

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

metrology METROLOGY

review metrology consistency

Keep the meaning precise: The science and practice of measurement.

product mix PRODUCT MIX

adjust for product mix

Keep the meaning precise: The composition of different products in a dataset.

B Precision and professional shorthand

EXTEND YOUR RANGE

SPC

Statistical process control, using data to monitor process behavior.

apply SPC

control chart

A time-ordered chart with a defined process-monitoring method.

review the control chart

control limit

A monitoring boundary derived under a stated process model or method.

distinguish control limits

specification limit

A defined product or process acceptance boundary.

check specification limits

common-cause variation

Variation inherent in the currently defined process system.

characterize common-cause variation

special-cause variation

Variation associated with a nonroutine identifiable influence.

investigate special-cause variation

control signal

A result meeting a defined rule for investigation on a control chart.

respond to a control signal

process stability

Consistent statistical behavior under the defined monitoring basis.

assess process stability

subgroup

A defined set of observations grouped for process analysis.

define rational subgroups

measurement system

The instruments, methods, people, and conditions producing measurements.

assess the measurement system

yield learning

Investigation that improves understanding of yield losses and drivers.

support yield learning

causal attribution

Assigning an observed effect to a particular cause.

qualify causal attribution

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

stratification STRATIFICATION

stratify by product

Keep the meaning precise: Separating data into relevant comparable groups.

common-cause variation COMMON-CAUSE VARIATION

characterize common-cause variation

Keep the meaning precise: Variation inherent in the currently defined process system.

C Phrases that move the work forward

Give the pilot counts
The pilot passed 92 of 100 tested die.

Give the reference counts
The reference passed 900 of 1,000 tested die.

Name the absolute change
The observed difference is two percentage points.

Name the relative basis
Relative to 90%, that is approximately 2.22%.

Expose the mix
The two lots have different product mixes.

Avoid causality by timing
The process change is not isolated by this comparison.

Qualify significance
Statistical significance has not been assessed.

Request like-for-like data
Compare relevant product groups on a consistent basis.

Separate limits
Control limits and specification limits serve different purposes.

Ask for time order
Which time-ordered evidence supports the stability claim?

Keep the method visible
State the subgrouping and monitoring rules.

Avoid dismissing a signal
A control signal warrants investigation under the defined procedure.

Avoid invented diagnosis
The signal does not identify its own cause.

Preserve useful progress
The pilot result is encouraging but limited.

Check measurement consistency
Confirm the measurement and test basis before combining data.

Close the finding
Report the counts, observed difference, limitations, and next analysis.

Precision in real exchanges

Purpose: Name the relative basis
Useful wording: Relative to 90%, that is approximately 2.22%.

Purpose: Expose the mix
Useful wording: The two lots have different product mixes.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which announcement is supported?

A. The process change is proven to improve every product by two relative percent.

B. The pilot proves the process is statistically stable.

C. Different denominators make both observed percentages meaningless.

D. The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.

2. What is the approximate relative increase from 90% to 92%?

A. 92.00%

B. 2.22%

C. 2.00%

D. 20.00%

3. A colleague needs the term for this meaning: The set of items included in a reported test result. Which term fits?

A. tested population

B. percentage point

C. subgroup

D. control limit

D Choose the response that fits / continued

4. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points." Ravi: [missing response] Elena: "I want the team to see the pilot as useful progress without overstating what caused the result. The change happened before the pilot, but timing is not enough." Which response belongs in the gap?

A. Use stratification to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

B. Check the measurement system and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.

C. Begin with the pass count and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.

D. Yes, use percentage point for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.

Answers and explanations, page 139

One useful distinction

The pilot has 92% yield and the reference 90%. The difference is 2 percentage points, or about 2.22% relative to 90%. Different product mixes and sample sizes mean this observation alone does not establish a causal improvement or statistical significance.

E Conversation

Two percentage points, not a proven process gain

Two percentage points, not a proven process gain

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

causal attribution / control signal / specification limit / percentage point / yield learning / process stability / product mix / measurement system / stratification / pass count

01 **Elena:** The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?

02 **Ravi:** Begin with the [1] _____ and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.

03 **Elena:** Put both counts beside the percentages. A reader needs to see that one result comes from a hundred die and the other from a thousand.

04 **Ravi:** Correct. Also state the [2] _____. The lots contain different product compositions, so the two percentages are not a controlled comparison of the process change alone.

05 **Elena:** The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points.

06 **Ravi:** Yes, use [3] _____ for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.

07 **Elena:** I want the team to see the pilot as useful progress without overstating what caused the result. The change happened before the pilot, but timing is not enough.

08 **Ravi:** Avoid unsupported [4] _____. The result can guide further investigation while the product mix and other relevant differences remain visible.

09 **Elena:** Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference.

10 **Ravi:** Use [5] _____ to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

Conversation continues on the next page.

Two percentage points, not a proven process gain

CONVERSATION / CONTINUED

11 **Elena:** The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries.

12 **Ravi:** Then remove that [6] _____ claim. A favorable summary percentage does not establish how the process behaves over time or whether a monitoring rule has signaled a change.

13 **Elena:** We have acceptance criteria for individual test results. Some colleagues are treating those limits as though they were the same as the chart's monitoring boundaries.

14 **Ravi:** A [7] _____ defines an acceptance boundary. A control limit has a different purpose, so passing the specification is not proof that the process is in statistical control.

15 **Elena:** If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it.

16 **Ravi:** Exactly. A [8] _____ is a prompt for the defined investigation process. Do not convert it directly into a root-cause statement or an automatic disposition for every unit.

17 **Elena:** We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change.

18 **Ravi:** Check the [9] _____ and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.

Two percentage points, not a proven process gain / continued

19 **Elena:** The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis.

20 **Ravi:** That supports [10] _____. The pilot remains informative, and the team gets a precise next question instead of a success claim that the present data cannot establish.

CHECK THE GAPS

1. pass count / **2.** product mix / **3.** percentage point / **4.** causal attribution / **5.** stratification / **6.** process stability / **7.** specification limit / **8.** control signal / **9.** measurement system / **10.** yield learning

Why these answers fit, page 141

F Check accuracy and scope

Two percentage points, not a proven process gain

Return to the case

Yield engineer Ravi presents a pilot lot with 92 passing die out of 100 tested, compared with a reference lot with 900 passing die out of 1,000 tested. The product mixes differ, and no controlled comparison or statistical significance assessment has been completed. Manager Elena wants to announce that a process change caused a two-percent yield improvement. The team also has no time-ordered control-chart analysis supporting a stability claim. Ravi must preserve the useful observations while correcting the causal, percentage, and stability language.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which next analysis addresses the product-mix problem?

A. Compare only the absolute pass counts without the tested populations.

B. Separate comparable product groups and review consistent test bases.

C. Pool both lots into one aggregate percentage and omit the product breakdown.

D. Equalize lot sample sizes while retaining their different product compositions.

2. Which statement distinguishes control and specification limits?

A. Control limits monitor behavior; specification limits define acceptance boundaries.

B. Control limits are customer acceptance boundaries; specification limits summarize observed variation.

C. A process within its specification limits necessarily has stable statistical behavior.

D. A process within its control limits necessarily meets all product requirements.

Answers and explanations, page 142

Listen for the distinction

Switch roles. Read turns 5-16, emphasizing percentage points, product mix, and the two types of limits.

Give the counts before the conclusion

A favorable yield difference can start an investigation without ending it. State the passing and tested counts, identify changed product mix, and distinguish an absolute percentage-point difference from a relative percentage change. Avoid using the words significant or stable as casual substitutes for better.

F Say it / transfer to a new case

Percentage points need a denominator

A trial passes 48 of 50 tested die, or 96%. A reference passes 450 of 500, or 90%. Product mixes differ, and no causal analysis is complete.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

percentage points / product mix / causal attribution / denominator

01 **Engineer:** Put 48 passing die over the trial's 50-die [1] _____.

02 **Reviewer:** Then show the changed [2] _____ beside the reference comparison, not in a separate appendix.

03 **Engineer:** I will report the difference as six [3] _____, not six relative percent.

04 **Reviewer:** Good. Leave the claim that the process caused it open until we have supported [4] _____.

CHECK THE GAPS

1. denominator / **2.** product mix / **3.** percentage points / **4.** causal attribution

Why these answers fit, page 142

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "We have acceptance criteria for individual test results. Some colleagues are treating those limits as though they were the same as the chart's monitoring boundaries." Ravi: [missing response] Elena: "If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it." Which response belongs in the gap?

A. Then remove that process stability claim. A favorable summary percentage does not establish how the process behaves over time or whether a monitoring rule has signaled a change.

B. Exactly. A control signal is a prompt for the defined investigation process. Do not convert it directly into a root-cause statement or an automatic disposition for every unit.

C. That supports yield learning. The pilot remains informative, and the team gets a precise next question instead of a success claim that the present data cannot establish.

D. A specification limit defines an acceptance boundary. A control limit has a different purpose, so passing the specification is not proof that the process is in statistical control.

Answers and explanations, page 143

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Deposition, Etch, CMP, and Process Windows" and "Metrology, SPC, and Yield Learning". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Chemical vapor deposition, forming a film through chemical reactions of vapor precursors. Which term fits?

A. overetch

B. nominal setting

C. step coverage

D. CVD

2. A colleague needs the term for this meaning: A lot used as a stated comparison basis. Which term fits?

A. metrology

B. SPC

C. control signal

D. reference lot

3. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement." Theo: [missing response] Lina: "The review also asks whether this is an alignment problem. Can we clarify that separately? I do not want the CD figure copied into an overlay conclusion." Which response belongs in the gap?

A. Request comparable remeasurement through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.

B. Exactly. Etch bias is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

C. Yes. Assess measurement bias and other relevant effects rather than force either outcome. The aim is a supported interpretation, not a preferred conclusion that the process changed or stayed identical.

D. The sampling site selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change." Ravi: [missing response] Elena: "The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis." Which response belongs in the gap?

A. Yes, use percentage point for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.

B. Use stratification to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

C. Begin with the pass count and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.

D. Check the measurement system and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.

Answers and explanations, page 167

A Read the situation

LESSON 05 / DEFECT DENSITY AND CLEANROOM CONTAMINATION

Describe inspection findings with location, area, timing, and classification while avoiding premature contamination-source claims.

Twenty-four detections are not a confirmed source

Inspection engineer Hana and contamination-control engineer Oscar review a note saying cleanroom problem. The underlying record contains 24 particle-like detections over 200 square centimeters inspected on wafers from lot P8 after tool C2 between 9 and 10 a.m. The map shows an edge cluster. No comparable pre-process scan exists, defect review is incomplete, and no source is confirmed. The lot is held under the local procedure. The team must distinguish detections, confirmed defects, added defects, and airborne particle measurements.

People: Hana, Inspection engineer; Oscar, Contamination-control engineer.

Read the record

Choose one answer per question using the facts above.

1. What does the inspection record contain?

A. 24 particle-like detections over 200 square centimeters
B. 24 confirmed electrical failures on every wafer
C. An airborne count per cubic meter
D. A measured increase of 24 defects from a matched pre-scan

2. What does the map show?

A. Uniform distribution across every layer
B. A confirmed cleanroom air source
C. A completed source elimination study
D. An edge cluster

3. Why cannot the 24 detections be called 24 added defects?

A. The lot has an identifier.
B. Surface maps always measure airborne particles.
C. No comparable pre-process scan exists, and review is incomplete.
D. The inspected area is known.

Answers and explanations, page 144

B Vocabulary in use

CORE TERMS

inspection detection
A feature flagged by an inspection method for assessment.

review inspection detections

defect review
Examination used to classify and interpret detected features.

complete defect review

defect density
Defects per unit area on a stated counting and inspection basis.

report defect density

inspected area
The surface area included in an inspection result.

state the inspected area

particle-like signature
An observed appearance consistent with a particle, pending review.

classify a particle-like signature

nuisance detection
A flagged feature not relevant to the defect concern being assessed.

filter nuisance detections

killer defect
A defect that causes functional failure in its particular context.

assess killer-defect relevance

defect classification
Assignment of detected features to defined categories.

refine defect classification

wafer map
A spatial representation of results on a wafer.

examine the wafer map

edge cluster
A concentration of observations near a wafer's edge.

investigate an edge cluster

spatial signature
A location pattern in measured or inspected results.

compare spatial signatures

added defect
A defect introduced between comparable inspection points.

quantify added defects

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

inspection detection INSPECTION DETECTION

review inspection detections

Keep the meaning precise: A feature flagged by an inspection method for assessment.

nuisance detection NUISANCE DETECTION

filter nuisance detections

Keep the meaning precise: A flagged feature not relevant to the defect concern being assessed.

B Precision and professional shorthand

EXTEND YOUR RANGE

pre-process scan

Inspection performed before a specified process step.

preserve the pre-process scan

post-process scan

Inspection performed after a specified process step.

compare post-process scans

airborne particle count

A count of particles in a defined air sample or volume.

review airborne particle counts

surface contamination

Unwanted material present on a surface.

investigate surface contamination

cross-contamination

Transfer of unwanted material between areas, items, or processes.

prevent cross-contamination

contamination source

The origin of unwanted material.

identify the contamination source

particle excursion

An unusual particle-related result under defined monitoring rules.

investigate a particle excursion

inspection sensitivity

The ability of an inspection method to detect specified features.

confirm inspection sensitivity

size threshold

The size boundary used for counting or classifying features.

state the size threshold

exposure history

The record of relevant environments and handling encountered.

reconstruct exposure history

suspect interval

A defined period being examined for possible affected material.

bound the suspect interval

containment scope

The material or operations covered by a temporary restriction.

define containment scope

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

added defect ADDED DEFECT

quantify added defects

Keep the meaning precise: A defect introduced between comparable inspection points.

cross-contamination CROSS-CONTAMINATION

prevent cross-contamination

Keep the meaning precise: Transfer of unwanted material between areas, items, or processes.

C Phrases that move the work forward

Replace the vague label
The record shows particle-like surface detections, not a confirmed cleanroom source.

Name the density basis
That is 0.12 detections per square centimeter.

Bound the interval
These records follow C2 between 9 and 10 a.m.

Avoid an added-defect claim
We lack a comparable pre-process scan.

Ask about sensitivity
Were the inspection settings and size threshold consistent?

Trace the material
Link the map to the lot, wafers, and process history.

Define the next review
Review the detections and relevant exposure history.

Ask about scope
Which material is included in the current containment?

Give the count and area
There were 24 detections over 200 square centimeters inspected.

Locate the pattern
The wafer map shows an edge cluster.

Preserve the uncertainty
Defect classification remains incomplete.

Separate measurement types
Surface detections are not airborne particle counts.

Avoid premature blame
The spatial signature does not confirm the source.

Keep the restriction clear
P8 remains held under the local procedure.

Avoid automatic yield claims
Not every flagged feature is a confirmed killer defect.

Close the note
Report observations, limits, containment status, and investigation ownership.

Precision in real exchanges

Purpose: Locate the pattern
Useful wording: The wafer map shows an edge cluster.

Purpose: Bound the interval
Useful wording: These records follow C2 between 9 and 10 a.m.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. What is the observed detection density?

A. 0.12 particles per cubic meter of air
B. 0.12 detections per square centimeter
C. 12 detections per square centimeter
D. 8.33 detections per square centimeter

2. Which conclusion is unsupported?

A. The records are after C2.
B. The map shows an edge cluster.
C. Defect review is incomplete.
D. C2 introduced all 24 detections.

3. A colleague needs the term for this meaning: The surface area included in an inspection result. Which term fits?

A. inspected area
B. airborne particle count
C. size threshold
D. wafer map

D Choose the response that fits / continued

4. Rebuild the exchange from "Twenty-four detections are not a confirmed source". Hana: "The map clusters near the edge. I will send the wafer IDs and coordinates so the team can examine the pattern without guessing its source." Oscar: [missing response] Hana: "The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan." Which response belongs in the gap?

A. Call it an edge cluster. Describe where it appears and connect the map to the wafer IDs, rather than turn the pattern into an immediate diagnosis of the cleanroom.

B. Start with the inspected area. The record covers two hundred square centimeters, so the number can be interpreted on a defined surface basis rather than as an unexplained total.

C. That keeps the contamination source question open on an accurate basis. We can investigate efficiently while preserving the distinction between an observation, a hypothesis, and a confirmed cause.

D. Exactly. A pre-process scan must be comparable with the later scan. Different sensitivity, locations, or thresholds could otherwise make a count difference look like process-added contamination.

Answers and explanations, page 144

One useful distinction

The observed detection density is 24 divided by 200, or 0.12 detections per square centimeter. Do not call it a confirmed killer-defect density. Classification is incomplete, and no comparable pre-scan establishes how many detections were introduced by C2.

E Conversation

Twenty-four detections are not a confirmed source

Twenty-four detections are not a confirmed source

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

inspected area / pre-process scan / exposure history / edge cluster / added defect / Defect review / airborne particle count / inspection sensitivity / contamination source / containment scope

01 **Hana:** The review note says cleanroom problem, but that label is much broader than the evidence. We have twenty-four particle-like detections on the inspected surfaces from P8.

02 **Oscar:** Start with the [1] _____. The record covers two hundred square centimeters, so the number can be interpreted on a defined surface basis rather than as an unexplained total.

03 **Hana:** That gives zero point one two detections per square centimeter. I should keep the word detections because the detailed classification has not been completed.

04 **Oscar:** Correct. [2] _____ is still open. A flagged feature is not automatically a confirmed defect of the relevant type, much less a demonstrated electrical failure.

05 **Hana:** The map clusters near the edge. I will send the wafer IDs and coordinates so the team can examine the pattern without guessing its source.

06 **Oscar:** Call it an [3] _____. Describe where it appears and connect the map to the wafer IDs, rather than turn the pattern into an immediate diagnosis of the cleanroom.

E Conversation / continued

07 **Hana:** The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan.

08 **Oscar:** We cannot claim an [4] _____ count on that basis. After C2 establishes the recorded sequence, not proof that C2 introduced every detected feature.

09 **Hana:** A matched earlier inspection would help distinguish what was already present from what appeared later. We would also need to check the inspection basis before comparing counts.

10 **Oscar:** Exactly. A [5] _____ must be comparable with the later scan. Different sensitivity, locations, or thresholds could otherwise make a count difference look like process-added contamination.

Conversation continues on the next page.

Twenty-four detections are not a confirmed source

CONVERSATION / CONTINUED

11 **Hana:** Facilities sent an air-monitoring result too. I will keep it separate from these surface counts; the units and measurement methods are different.

12 **Oscar:** An [6] _____ is a different measurement. A value per air volume cannot be treated as the same quantity as detections per inspected surface area.

13 **Hana:** I will include the inspection recipe and threshold with the maps. That will let reviewers see whether the method was capable of detecting the features being discussed.

14 **Oscar:** Keep [7] _____ visible. A tool finding more features after a settings change does not by itself prove that the process produced more contamination.

15 **Hana:** P8 remains on hold under the local procedure. The investigation team needs to say which material and time interval are included, rather than leave the restriction implicit.

16 **Oscar:** Define the [8] _____ through the authorized process. A precise scope supports coordination without turning this discussion into permission to release or extend the hold arbitrarily.

17 **Hana:** We can reconstruct the lot's recent handling and process history alongside the spatial pattern. That should help distinguish possible explanations and identify useful follow-up evidence.

18 **Oscar:** Review the [9] _____. It can guide source investigation, but a shared tool or time interval is not enough on its own to establish the origin.

19 **Hana:** The corrected note will give the count, area, edge pattern, interval, and incomplete classification. It will state that introduction by C2 and an airborne source are unconfirmed.

20 **Oscar:** That keeps the [10] _____ question open on an accurate basis. We can investigate efficiently while preserving the distinction between an observation, a hypothesis, and a confirmed cause.

CHECK THE GAPS

1. inspected area / **2.** Defect review / **3.** edge cluster / **4.** added defect / **5.** pre-process scan / **6.** airborne particle count / **7.** inspection sensitivity / **8.** containment scope / **9.** exposure history / **10.** contamination source

Why these answers fit, page 146

F Check accuracy and scope

Twenty-four detections are not a confirmed source

Return to the case

Inspection engineer Hana and contamination-control engineer Oscar review a note saying cleanroom problem. The underlying record contains 24 particle-like detections over 200 square centimeters inspected on wafers from lot P8 after tool C2 between 9 and 10 a.m. The map shows an edge cluster. No comparable pre-process scan exists, defect review is incomplete, and no source is confirmed. The lot is held under the local procedure. The team must distinguish detections, confirmed defects, added defects, and airborne particle measurements.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which next record is useful for the investigation?

A. A whole-fab monthly total used to assign the source for this lot
B. A surface map converted to an airborne count without an air sample
C. Traceable wafer, inspection-setting, timing, and exposure information
D. A count from another layer using a different threshold, treated as a matched pre-scan

2. Which distinction is correct?

A. Surface detection density and airborne particle counts use different measurement bases.
B. A post-process detection count equals an added-defect count without a pre-scan.
C. An inspection's flagged-feature count is already the confirmed killer-defect count.
D. An edge pattern identifies the contamination source without further review.

Answers and explanations, page 147

Listen for the distinction

Switch roles for turns 5-14. Keep the edge pattern, possible source, and surface-versus-air distinction explicit.

Describe the pattern before naming the source

A location pattern can focus an investigation without proving where contamination originated. State the observation and its limits, then request traceable process and inspection records. Avoid turning a vague cleanroom label into blame for a team, operator, or tool before the evidence supports it.

F Say it / transfer to a new case

Count the detections, not an invented cause

A post-process scan flags 12 features over 100 square centimeters. Classification is pending, and no comparable pre-process scan exists. No contamination source is confirmed.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

inspected area / pre-process scan / defect review / contamination source

01 **Engineer:** Keep 100 square centimeters beside the count as the [1] _____.

02 **Reviewer:** And leave the twelve flagged features unclassified pending [2] _____.

03 **Engineer:** I will request a comparable [3] _____ before reporting any process-added count.

04 **Reviewer:** Thank you. Do not name the last tool as the [4] _____ without supporting investigation.

CHECK THE GAPS

1. inspected area / **2.** defect review / **3.** pre-process scan / **4.** contamination source

Why these answers fit, page 147

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Twenty-four detections are not a confirmed source". Hana: "I will include the inspection recipe and threshold with the maps. That will let reviewers see whether the method was capable of detecting the features being discussed." Oscar: [missing response] Hana: "P8 remains on hold under the local procedure. The investigation team needs to say which material and time interval are included, rather than leave the restriction implicit." Which response belongs in the gap?

A. Keep inspection sensitivity visible. A tool finding more features after a settings change does not by itself prove that the process produced more contamination.

B. Exactly. A pre-process scan must be comparable with the later scan. Different sensitivity, locations, or thresholds could otherwise make a count difference look like process-added contamination.

C. We cannot claim an added defect count on that basis. After C2 establishes the recorded sequence, not proof that C2 introduced every detected feature.

D. Correct. Defect review is still open. A flagged feature is not automatically a confirmed defect of the relevant type, much less a demonstrated electrical failure.

Answers and explanations, page 148

A Read the situation

LESSON 06 / EQUIPMENT UPTIME, RECIPES, AND TOOL MATCHING

Distinguish recipe identity, specification performance, tool matching, and equipment availability in a transfer discussion.

Same recipe, different means

Equipment engineer Victor and process engineer Asha compare tools A and B using the same identified R7 recipe and an agreed comparable measurement basis. Their film-thickness sample means are 98 nm and 102 nm. The local mean specification is 95 to 105 nm, but the tool-matching criterion requires the means to differ by no more than 2 nm. Tool B was available for 900 of a defined 1,000-minute window. A colleague uses the identical recipe and 90% availability to claim B is an approved interchangeable backup. No transfer approval is recorded.

People: Victor, Equipment engineer; Asha, Process engineer.

Read the record

Choose one answer per question using the facts above.

1. How far apart are the sample means?

A. 2 nm
B. 7 nm
C. 200 nm
D. 4 nm

2. Do the means meet the stated matching criterion?

A. Yes; both tools use R7.
B. The availability percentage automatically settles matching.
C. No; 4 nm exceeds the allowed 2 nm difference.
D. Yes; both means are within the mean specification.

3. What does 90% describe here?

A. Availability for 900 of the defined 1,000 minutes
B. The proportion of wafers passing final test
C. A 90% probability that the tools are interchangeable
D. A recorded production transfer approval

Answers and explanations, page 149

B Vocabulary in use

CORE TERMS

tool matching
Demonstrating comparable tool performance against defined criteria.

assess tool matching

chamber matching
Demonstrating comparable performance between process chambers.

verify chamber matching

recipe revision
The identified version of a process recipe.

confirm the recipe revision

setpoint
A commanded target value for a controlled variable.

record the setpoint

actual value
The measured or achieved value rather than the commanded target.

compare setpoint and actual value

equipment state
The recorded operational condition of a tool.

report equipment state

availability
The share of a defined time basis during which equipment is available.

calculate equipment availability

utilization
The share of a stated time basis during which equipment is used.

state the utilization denominator

scheduled downtime
Planned time when equipment is unavailable for the defined use.

plan scheduled downtime

unscheduled downtime
Unplanned time when equipment is unavailable.

investigate unscheduled downtime

preventive maintenance
Planned work intended to reduce equipment failures or deterioration.

schedule preventive maintenance

corrective maintenance
Work to restore equipment after a fault or failure.

complete corrective maintenance

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

tool matching TOOL MATCHING

assess tool matching

Keep the meaning precise: Demonstrating comparable tool performance against defined criteria.

equipment state EQUIPMENT STATE

report equipment state

Keep the meaning precise: The recorded operational condition of a tool.

B Precision and professional shorthand

EXTEND YOUR RANGE

MTBF

Mean time between failures under a stated operating and event definition.

report MTBF

MTTR

Mean time to repair or restore under the stated local definition.

define the MTTR basis

chamber condition

The physical and operating state of a process chamber.

assess chamber condition

seasoning

Controlled processing used to establish a required chamber condition.

verify seasoning status

calibration status

The current documented state of required measurement calibration.

check calibration status

qualification wafer

A wafer used to assess a specified tool or process condition.

evaluate a qualification wafer

monitor wafer

A wafer used to track selected process or tool behavior.

review monitor-wafer results

matching criterion

A defined limit for acceptable performance difference.

apply the matching criterion

process transfer

Moving a defined process to another tool or location.

approve process transfer

backup tool

An alternative tool intended for use under defined conditions.

qualify a backup tool

release status

Whether the required authorization for a specified use is in place.

confirm release status

offset correction

A controlled adjustment addressing a measured systematic difference.

assess offset correction

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

corrective maintenance CORRECTIVE MAINTENANCE

complete corrective maintenance

Keep the meaning precise: Work to restore equipment after a fault or failure.

calibration status CALIBRATION STATUS

check calibration status

Keep the meaning precise: The current documented state of required measurement calibration.

C Phrases that move the work forward

Confirm the recipe
Both tools used the identified R7 revision.

Separate inputs from outputs
Identical setpoints do not prove identical performance.

State the means
A measured 98 nm and B measured 102 nm.

Give the difference
The means are 4 nm apart.

Apply the right criterion
Matching allows no more than a 2 nm difference.

Separate the specifications
Both means are in spec, but the pair is not matched.

Bound the evidence
These are sample means, not every individual wafer result.

Ask about chamber state
Were the relevant chamber conditions documented?

Check measurement status
Confirm the calibration and measurement basis.

State availability
B was available for 900 of the defined 1,000 minutes.

Keep the percentage limited
That 90% figure is availability, not yield.

Avoid automatic substitution
Backup status needs the required qualification and approval.

Route the investigation
Review the observed offset before proposing a correction.

Preserve authorization
Do not change the recipe from this comparison alone.

Clarify the transfer
Which process, product, and tool combination is proposed?

Close the status
Report specification, matching, availability, and release separately.

Precision in real exchanges

Purpose: Give the difference
Useful wording: The means are 4 nm apart.

Purpose: Apply the right criterion
Useful wording: Matching allows no more than a 2 nm difference.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which status report is accurate?

A. Tool B is approved because its availability is 90%.

B. Every wafer from both tools is proven in specification.

C. Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.

D. Both tools are matched because 98 and 102 lie within 95 to 105.

2. What does using the same recipe establish?

A. A matching recipe revision substitutes for product-specific transfer approval.

B. The identified recipe input is shared, not that output performance is interchangeable.

C. The two tools have equivalent physical chamber conditions throughout the run.

D. The same nominal setpoints remove the need to compare actual results.

3. A colleague needs the term for this meaning: A commanded target value for a controlled variable. Which term fits?

A. setpoint

B. scheduled downtime

C. chamber condition

D. process transfer

D Choose the response that fits / continued

4. Rebuild the exchange from "Same recipe, different means". Victor: "Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed." Asha: [missing response] Victor: "I will say both sample means meet that limit. We have not reviewed every individual measurement, so I cannot extend that claim to every wafer." Which response belongs in the gap?

A. Label it availability and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.

B. Correct. The equipment state record helps explain the time measure, but an available tool is not automatically qualified for this product and process combination.

C. Check calibration status through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.

D. The actual value has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.

Answers and explanations, page 149

> **One useful distinction**
>
> Both sample means meet the local 95-to-105 nm mean specification. Their difference is 4 nm, so they fail the separate 2 nm matching criterion. This does not establish the status of every individual wafer, whose measurements are not supplied.

E Conversation

Same recipe, different means

Same recipe, different means

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

actual value / release status / recipe revision / availability / process transfer / equipment state / chamber condition / matching criterion / calibration status / Tool matching

01 **Victor:** Tool B has the same R7 recipe as A and was available for ninety percent of the window. The planning note calls it an interchangeable backup.

02 **Asha:** That combines different questions. [1] _____ depends on the measured outcomes and our criterion, not only the recipe name or the amount of time the tool was available.

03 **Victor:** We confirmed the same recipe revision and an agreed measurement basis. A's sample mean was ninety-eight nanometers, while B's was one hundred two.

04 **Asha:** The [2] _____ allows a difference of no more than two nanometers. These means differ by four, so this comparison does not establish an acceptable match.

05 **Victor:** Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed.

06 **Asha:** The [3] _____ has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.

E Conversation / continued

07 **Victor:** I will say both sample means meet that limit. We have not reviewed every individual measurement, so I cannot extend that claim to every wafer.

08 **Asha:** Agreed. Confirm the [4] _____ and report the evidence at its actual level. Do not let a summary mean or a shared input become a claim about every processed unit.

09 **Victor:** The observed difference needs investigation. Chamber history and maintenance condition may be relevant, but neither has been established as the cause from these numbers.

10 **Asha:** Review [5] _____ and the other appropriate records. Identical commanded settings do not guarantee identical physical conditions or measurement behavior.

Conversation continues on the next page.

Same recipe, different means

CONVERSATION / CONTINUED

11 **Victor:** I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?

12 **Asha:** Check [6] _____ through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.

13 **Victor:** For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent.

14 **Asha:** Label it [7] _____ and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.

15 **Victor:** We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions.

16 **Asha:** Correct. The [8] _____ record helps explain the time measure, but an available tool is not automatically qualified for this product and process combination.

17 **Victor:** The backup proposal remains useful because capacity is constrained, but it needs the appropriate technical assessment and approval before production relies on it.

18 **Asha:** Treat it as a [9] _____ decision. The defined scope, matching evidence, and required authorization must be clear instead of being inferred from the existence of an alternative tool.

19 **Victor:** I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable.

20 **Asha:** That preserves the [10] _____ accurately. We can pursue the backup option while making its unresolved requirements visible to the people planning production.

CHECK THE GAPS

1. Tool matching / **2.** matching criterion / **3.** actual value / **4.** recipe revision / **5.** chamber condition / **6.** calibration status / **7.** availability / **8.** equipment state / **9.** process transfer / **10.** release status

Why these answers fit, page 150

F Check accuracy and scope

Same recipe, different means

Return to the case

Equipment engineer Victor and process engineer Asha compare tools A and B using the same identified R7 recipe and an agreed comparable measurement basis. Their film-thickness sample means are 98 nm and 102 nm. The local mean specification is 95 to 105 nm, but the tool-matching criterion requires the means to differ by no more than 2 nm. Tool B was available for 900 of a defined 1,000-minute window. A colleague uses the identical recipe and 90% availability to claim B is an approved interchangeable backup. No transfer approval is recorded.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which calculation matches the availability basis?

A. 100 unavailable minutes divided by 1,000 gives 10% availability.
B. The available-time count alone establishes utilization without a used-time count.
C. 900 divided by 1,000 equals 90%.
D. 900 divided by the 100 unavailable minutes equals the availability fraction.

2. What remains necessary before calling B an approved interchangeable backup?

A. Only a passing sample mean against the broad mean specification
B. The required matching or qualification evidence and transfer approval
C. Only a repeated statement that R7 is installed
D. Only a higher availability percentage

Answers and explanations, page 152

Listen for the distinction

Switch roles. Read the availability exchange without replacing availability with yield or utilization.

Name which readiness question you answered

Available, in specification, matched, and released describe different conditions. A tool may satisfy one and not another. Give each claim its own evidence and criterion so production planning does not convert equipment uptime into process-transfer approval.

F Say it / transfer to a new case

Two different limits

Two tools have sample means of 50 and 53 nm. Both meet a mean specification of 48 to 55 nm. Matching allows at most 1 nm difference. Transfer approval is pending.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

specification / process transfer / release status / matching criterion

01 **Engineer:** Both means meet the 48-to-55 nm [1] _____.

02 **Reviewer:** But their three-nanometer gap misses our one-nanometer [2] _____.

03 **Engineer:** Then moving this process to the other tool remains a proposed [3] _____.

04 **Reviewer:** Correct. Keep approval pending in the [4] _____ field.

CHECK THE GAPS

1. specification / **2.** matching criterion / **3.** process transfer / **4.** release status

Why these answers fit, page 152

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Same recipe, different means". Victor: "For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent." Asha: [missing response] Victor: "We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions." Which response belongs in the gap?

A. The actual value has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.

B. Check calibration status through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.

C. The matching criterion allows a difference of no more than two nanometers. These means differ by four, so this comparison does not establish an acceptable match.

D. Label it availability and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.

Answers and explanations, page 153

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Defect Density and Cleanroom Contamination" and "Equipment Uptime, Recipes, and Tool Matching". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Defects per unit area on a stated counting and inspection basis. Which term fits?

A. post-process scan
B. defect density
C. defect classification
D. inspection sensitivity

2. A colleague needs the term for this meaning: The share of a stated time basis during which equipment is used. Which term fits?

A. tool matching
B. monitor wafer
C. utilization
D. MTBF

3. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference." Ravi: [missing response] Elena: "The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries." Which response belongs in the gap?

A. Check the measurement system and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.
B. Yes, use percentage point for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.
C. Begin with the pass count and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.
D. Use stratification to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "Same recipe, different means". Victor: "The backup proposal remains useful because capacity is constrained, but it needs the appropriate technical assessment and approval before production relies on it." Asha: [missing response] Victor: "I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable." Which response belongs in the gap?

A. That combines different questions. Tool matching depends on the measured outcomes and our criterion, not only the recipe name or the amount of time the tool was available.

B. The matching criterion allows a difference of no more than two nanometers. These means differ by four, so this comparison does not establish an acceptable match.

C. Treat it as a process transfer decision. The defined scope, matching evidence, and required authorization must be clear instead of being inferred from the existence of an alternative tool.

D. Agreed. Confirm the recipe revision and report the evidence at its actual level. Do not let a summary mean or a shared input become a claim about every processed unit.

Answers and explanations, page 170

A Read the situation

LESSON 07 / PACKAGING, TEST, AND RELIABILITY QUALIFICATION

Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.

One completed test is not complete qualification

Product engineer Diego and reliability engineer Mei review package Q7. Its approved fictional qualification plan requires temperature cycling, high-temperature operating life testing, a highly accelerated stress test, and final review of the evidence. Temperature cycling has passed under its protocol. Operating-life testing is running, and the accelerated stress test has not started. No final approval or accepted substitution is recorded. A customer slide nevertheless says qualification complete. The team must replace that headline with a precise status without describing open tests as failures.

People: Diego, Product engineer; Mei, Reliability engineer.

Read the record

Choose one answer per question using the facts above.

1. Which test has passed?

A. Temperature cycling
B. High-temperature operating life
C. The highly accelerated stress test
D. Every required test and final review

2. What is the operating-life test status?

A. Running
B. Passed and approved
C. Not required by the plan
D. Replaced by an accepted substitution

3. Why is qualification complete unsupported?

A. Every package test is optional.
B. A running test is automatically a failure.
C. The package identifier proves approval.
D. Two required tests and final review remain open.

Answers and explanations, page 154

B Vocabulary in use

CORE TERMS

package substrate
The package structure carrying connections between die and external interfaces.

specify the package substrate

flip-chip attachment
Connecting a die face-down through conductive bumps or pillars.

evaluate flip-chip attachment

mold compound
Encapsulating material used in many semiconductor packages.

qualify the mold compound

BGA
Ball grid array, a package connection format using an array of solder balls.

assess BGA assembly

SiP
System in package, integrating multiple functions or components in one package.

evaluate a SiP design

package construction
The materials and structural arrangement of a package.

identify package construction

wire bonding
Forming electrical connections using fine wires.

qualify wire bonding

underfill
Material placed around interconnections to support a package assembly.

assess underfill integrity

solder joint
A soldered electrical and mechanical connection.

inspect solder joints

QFN
Quad flat no-lead, a package format with exposed connection lands.

review the QFN package

TSV
Through-silicon via, a conductive connection passing through silicon.

characterize TSV connections

moisture sensitivity
Susceptibility to moisture-related effects under stated handling or stress conditions.

assess moisture sensitivity

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

package substrate PACKAGE SUBSTRATE

specify the package substrate

Keep the meaning precise: The package structure carrying connections between die and external interfaces.

solder joint SOLDER JOINT

inspect solder joints

Keep the meaning precise: A soldered electrical and mechanical connection.

B Precision and professional shorthand

EXTEND YOUR RANGE

delamination
Separation between bonded material layers.
inspect for delamination

thermal cycling
Repeated temperature changes under a specified test profile.
complete thermal cycling

HTOL
High-temperature operating life, an operating stress test under defined conditions.
report HTOL status

HAST
Highly accelerated stress test, a defined accelerated humidity-related stress test.
schedule HAST

preconditioning
Defined preparation before a test to represent relevant handling or assembly stresses.
document preconditioning

qualification plan
The defined tests, evidence, scope, and approval requirements for qualification.
follow the qualification plan

qualification vehicle
The selected device or structure used to support qualification evidence.
identify the qualification vehicle

test coverage
The requirements or failure concerns addressed by the performed tests.
assess qualification test coverage

failure analysis
Investigation of failed samples and their causes.
initiate failure analysis

qualification by similarity
Use of justified related evidence under an accepted qualification approach.
assess qualification by similarity

final test
Electrical testing at the defined final product stage.
complete final test

qualification approval
Authorized acceptance of the required qualification evidence.
obtain qualification approval

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

moisture sensitivity MOISTURE SENSITIVITY

assess moisture sensitivity

Keep the meaning precise: Susceptibility to moisture-related effects under stated handling or stress conditions.

preconditioning PRECONDITIONING

document preconditioning

Keep the meaning precise: Defined preparation before a test to represent relevant handling or assembly stresses.

C Phrases that move the work forward

Name the completed result

Q7 passed temperature cycling under the stated protocol.

State ongoing work

HTOL is still running.

State unstarted work

HAST has not started.

Keep the plan visible

The approved plan requires all three tests and final review.

Correct the headline

Qualification is in progress, not complete.

Avoid treating pending as failed

No result is available for the unstarted test.

Preserve configuration

Which package construction does the evidence cover?

Ask about the vehicle

Is this the approved qualification vehicle?

Check preparation

Are the required preconditioning records included?

Separate production test

Final test is not a substitute for every reliability stress.

Qualify related evidence

Similarity needs an accepted technical rationale.

Avoid a schedule shortcut

A desired customer date does not remove a required test.

State the approval gap

Final qualification approval is not recorded.

Describe coverage

Identify what each completed test actually addresses.

Commit to the next update

The next report will distinguish completed, running, and unstarted tests.

Close accurately

Keep release decisions with the authorized review process.

Precision in real exchanges

Purpose: Keep the plan visible
Useful wording: The approved plan requires all three tests and final review.

Purpose: Correct the headline
Useful wording: Qualification is in progress, not complete.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which customer headline is supported?

A. Q7 is fully qualified because temperature cycling passed.
B. Q7 failed HAST because HAST has not started.
C. Q7 is approved by similarity despite no accepted substitution.
D. Qualification in progress: temperature cycling passed; HTOL running; HAST not started.

2. What does the passing temperature-cycle result establish?

A. Completion of the operating-life test because both involve temperature
B. Completion of HAST because both are accelerated stress tests
C. Qualification of every package variant sharing the Q7 marketing name
D. The stated test result under its protocol and scope

3. A colleague needs the term for this meaning: Material placed around interconnections to support a package assembly. Which term fits?

A. SiP
B. HTOL
C. underfill
D. failure analysis

D Choose the response that fits / continued

4. Rebuild the exchange from "One completed test is not complete qualification". Diego: "The operating-life test is still running. The lab has not supplied its final result, so the customer should not read the current status as a pass." Mei: [missing response] Diego: "The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result." Which response belongs in the gap?

A. Label HTOL as running. If a progress checkpoint is available, identify it as interim information without treating it as the completed test outcome.

B. That preserves qualification approval as a separate decision. The next update can show genuine progress while keeping the unresolved tests and final review clear.

C. Exactly. HAST remains unstarted in this plan. Use that status rather than either a reassuring green result or an unsupported failure label.

D. Keep the package construction linked to the evidence. Qualification scope should not silently expand to a different variant simply because the marketing name is similar.

Answers and explanations, page 154

> **One useful distinction**
>
> This case uses a specific fictional plan, not a universal mandatory three-test list. One passing test supports its defined result. It does not complete the two open tests or the final review, and it does not establish every application-specific reliability claim.

E Conversation

One completed test is not complete qualification

One completed test is not complete qualification

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

qualification vehicle / preconditioning / HAST / package construction / qualification approval / HTOL / Test coverage / Qualification by similarity / qualification plan / thermal cycling

01 **Diego:** The customer slide says Q7 qualification is complete because temperature cycling passed. We still have two other tests in the plan, so I need a more accurate headline.

02 **Mei:** Start from the [1] _____. Our approved plan requires all three tests and final evidence review. We have completed one part, not the entire qualification decision.

03 **Diego:** The completed test followed its protocol and passed. I want to preserve that positive result rather than make the update sound as though nothing has been achieved.

04 **Mei:** Report the [2] _____ result clearly. Its value is not diminished by stating the other work accurately, and it should not be expanded beyond its protocol and scope.

05 **Diego:** The operating-life test is still running. The lab has not supplied its final result, so the customer should not read the current status as a pass.

06 **Mei:** Label [3] _____ as running. If a progress checkpoint is available, identify it as interim information without treating it as the completed test outcome.

07 **Diego:** The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result.

08 **Mei:** Exactly. [4] _____ remains unstarted in this plan. Use that status rather than either a reassuring green result or an unsupported failure label.

09 **Diego:** Which package construction should I name on the slide? Q7 alone may cover variants with different materials or assembly details.

10 **Mei:** Keep the [5] _____ linked to the evidence. Qualification scope should not silently expand to a different variant simply because the marketing name is similar.

Conversation continues on the next page.

One completed test is not complete qualification

CONVERSATION / CONTINUED

11 **Diego:** We should confirm the tested device or structure as well. Otherwise reviewers may not know how the selected samples relate to the configuration being discussed.

12 **Mei:** Identify the [6] _____ and its approved rationale. Traceability makes it possible to review applicability rather than assume that any available sample represents every product.

13 **Diego:** The assembly preparation records also belong in the package. I do not want the final review to discover that important preparation details were omitted from the summary.

14 **Mei:** Include the required [7] _____ records where the plan calls for them. A passing result needs the relevant preparation and test context to be interpreted correctly.

15 **Diego:** Sales wants to reuse results from an older package. I will refer that request for review; we have no accepted substitution for the open tests.

16 **Mei:** [8] _____ needs an accepted basis under the applicable process. A related package or a desired delivery date does not automatically change the approved requirements.

17 **Diego:** We also have final electrical test records, but those should not be presented as though they address every stress-related concern in the qualification plan.

18 **Mei:** Correct. [9] _____ differs between production checks and reliability assessments. Each result needs to be connected to what it actually evaluates, not treated as interchangeable evidence.

19 **Diego:** I will change the headline to qualification in progress, list the three statuses, and keep the configuration and outstanding review visible. We will not announce release from this slide.

20 **Mei:** That preserves [10] _____ as a separate decision. The next update can show genuine progress while keeping the unresolved tests and final review clear.

CHECK THE GAPS

1. qualification plan / **2.** thermal cycling / **3.** HTOL / **4.** HAST / **5.** package construction / **6.** qualification vehicle / **7.** preconditioning / **8.** Qualification by similarity / **9.** Test coverage / **10.** qualification approval

Why these answers fit, page 156

F Check accuracy and scope

One completed test is not complete qualification

Return to the case

Product engineer Diego and reliability engineer Mei review package Q7. Its approved fictional qualification plan requires temperature cycling, high-temperature operating life testing, a highly accelerated stress test, and final review of the evidence. Temperature cycling has passed under its protocol. Operating-life testing is running, and the accelerated stress test has not started. No final approval or accepted substitution is recorded. A customer slide nevertheless says qualification complete. The team must replace that headline with a precise status without describing open tests as failures.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. When could related evidence replace a required test in this case?

A. When the related package has the same outside dimensions, without a construction review
B. When the older report is newer than the current plan, without an applicability assessment
C. When final electrical test passes, without assessing the reliability evidence gap
D. Only through the applicable accepted change or similarity process, which has not occurred

2. What should final review examine?

A. Required evidence, scope, configuration, open items, and approval conditions
B. Only whether the slide uses the word complete
C. Only the date the first test started
D. A presumed pass for all tests without final results

Answers and explanations, page 157

Listen for the distinction

Switch roles for turns 9-20. Keep construction, preparation, related evidence, and final approval distinct.

Give progress without collapsing the plan

A customer may prefer a short headline, but complete must refer to the required plan and configuration. Name the finished test, the open work, and the pending decision. Do not imply that a test still running has passed, or that a test not yet started has failed.

F Say it / transfer to a new case

Pending is neither passed nor failed

A fictional package plan requires tests A, B, and C plus final review. A passed, B is running, and C has not started. No substitution or final approval is recorded.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

approval / rationale / evidence / plan

01 **Engineer:** Keep A's passing result as test [1] _____, not a claim that every requirement is complete.

02 **Reviewer:** B and C are still required by the approved [2] _____.

03 **Engineer:** We would need an accepted technical [3] _____ before substituting related results.

04 **Reviewer:** Until the tests and final review are complete, do not announce qualification [4] _____.

CHECK THE GAPS

1. evidence / **2.** plan / **3.** rationale / **4.** approval

Why these answers fit, page 157

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "One completed test is not complete qualification". Diego: "The assembly preparation records also belong in the package. I do not want the final review to discover that important preparation details were omitted from the summary." Mei: [missing response] Diego: "Sales wants to reuse results from an older package. I will refer that request for review; we have no accepted substitution for the open tests." Which response belongs in the gap?

A. Identify the qualification vehicle and its approved rationale. Traceability makes it possible to review applicability rather than assume that any available sample represents every product.

B. Include the required preconditioning records where the plan calls for them. A passing result needs the relevant preparation and test context to be interpreted correctly.

C. Keep the package construction linked to the evidence. Qualification scope should not silently expand to a different variant simply because the marketing name is similar.

D. Start from the qualification plan. Our approved plan requires all three tests and final evidence review. We have completed one part, not the entire qualification decision.

Answers and explanations, page 158

A Read the situation

LESSON 08 / FOUNDRY, TAPE-OUT, PDK, AND CAPACITY COMMUNICATION

Negotiate a requested date by separating design sign-off, data delivery, foundry acceptance, and capacity confirmation.

A possible slot is not tape-out readiness

Customer design lead Sofia asks foundry liaison Ken to move a planned October 15 tape-out to October 8. Four design-rule-check findings and one layout-versus-schematic mismatch remain unresolved in the agreed fictional PDK revision P3. The foundry has mentioned a possible earlier slot but has not reserved or accepted it. No complete sign-off package is available, and no waiver is approved. Ken must communicate the blockers and conditional next steps without promising a wafer start, finished samples, or shipment.

People: Sofia, Customer design lead; Ken, Foundry liaison.

Read the record

Choose one answer per question using the facts above.

1. Which earlier date is requested?

A. October 15 completed wafer sort
B. October 8 tape-out
C. October 8 finished samples
D. October 15 shipment

2. What remains unresolved in the design checks?

A. An approved waiver for all findings
B. Four DRC findings and one LVS mismatch
C. A completed sign-off package with no open items
D. Only a customer shipping address

3. What is the capacity status?

A. A confirmed wafer start
B. A guaranteed sample delivery
C. An approved bypass of design sign-off
D. A possible earlier slot, not reserved or accepted

Answers and explanations, page 159

B Vocabulary in use

CORE TERMS

foundry
A manufacturer producing semiconductor devices for other companies.

coordinate with the foundry

fabless company
A chip-design company that contracts out fabrication.

support a fabless customer

IDM
Integrated device manufacturer, combining chip design and manufacturing.

work with an IDM

PDK
Process design kit: process-specific models, rules, and design resources.

confirm the PDK revision

design rule
A process-specific constraint governing layout features or relationships.

check design rules

DRC
Design rule checking, testing layout against applicable rule definitions.

resolve DRC findings

LVS
Layout versus schematic checking, comparing extracted layout connectivity with the schematic.

close an LVS mismatch

rule deck
The implemented rules used by a physical-verification tool.

identify the rule deck

sign-off
The required formal completion and approval of defined checks.

complete design sign-off

tape-out
Release of the approved design dataset for the intended fabrication handoff.

plan the tape-out

GDSII
A widely used layout-data format for integrated circuit design.

deliver a GDSII dataset

OASIS
A compact integrated-circuit layout interchange format.

verify the OASIS file

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

foundry FOUNDRY

coordinate with the foundry

Keep the meaning precise: A manufacturer producing semiconductor devices for other companies.

DRC DRC

resolve DRC findings

Keep the meaning precise: Design rule checking, testing layout against applicable rule definitions.

B Precision and professional shorthand

EXTEND YOUR RANGE

mask data preparation

Processing layout data for mask manufacture.

schedule mask data preparation

mask set

The collection of masks needed for a defined fabrication flow.

release the mask set

engineering change order

A controlled record authorizing a defined engineering change, abbreviated ECO.

approve an engineering change order

waiver

An authorized exception to a specific requirement under an applicable process.

request a defined waiver

data acceptance

Confirmation that a delivered dataset meets the receiving process's requirements.

obtain foundry data acceptance

capacity reservation

A confirmed allocation of manufacturing capacity.

confirm capacity reservation

wafer start

Entry of a wafer lot into the defined fabrication flow.

confirm the wafer start

cycle time

Elapsed time through a defined manufacturing flow or stage.

state fabrication cycle time

engineering lot

A lot used for development or engineering evaluation.

plan an engineering lot

risk production

Production begun before all defined qualification or maturity conditions are complete.

assess risk-production approval

respin

A revised design iteration requiring another fabrication cycle.

evaluate the need for a respin

delivery commitment

An agreed promise about a specified deliverable and date.

qualify the delivery commitment

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

OASIS OASIS

verify the OASIS file

Keep the meaning precise: A compact integrated-circuit layout interchange format.

data acceptance DATA ACCEPTANCE

obtain foundry data acceptance

Keep the meaning precise: Confirmation that a delivered dataset meets the receiving process's requirements.

C Phrases that move the work forward

Name the requested milestone

You are requesting October 8 tape-out.

State the open checks

Four DRC findings and one LVS mismatch remain open.

Avoid silent version changes

Keep the rule deck and design dataset identified.

Ask about acceptance

What is required for the foundry to accept the dataset?

Bound the exception

No waiver is approved for these findings.

Request a feasible sequence

Align check closure, data review, and capacity confirmation.

Clarify ownership

Who owns each open check and the next status update?

State the next update

We will report check status and slot confirmation separately.

Separate later dates

That is not a wafer-start or sample-delivery commitment.

Confirm the basis

The checks use the agreed PDK revision P3.

Qualify the slot

The foundry mentioned a possibility, not a reservation.

Keep sign-off visible

The complete sign-off package is not yet available.

Preserve the decision route

Any exception must follow the applicable authorization process.

Avoid a downstream promise

We cannot infer finished samples from a tape-out target.

Record the dependency

The earlier request depends on both technical readiness and capacity.

Close conditionally

October 8 remains a request pending the required evidence and acceptance.

Precision in real exchanges

Purpose: Confirm the basis
Useful wording: The checks use the agreed PDK revision P3.

Purpose: Avoid silent version changes
Useful wording: Keep the rule deck and design dataset identified.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which reply to the earlier-date request is supported?

A. October 8 remains conditional on check closure, the required handoff, and capacity confirmation.

B. October 8 guarantees finished samples because a slot was mentioned.

C. The unresolved findings disappear if the customer accepts schedule risk.

D. The foundry has already reserved capacity by discussing it.

2. What does an LVS mismatch concern?

A. A difference between extracted layout connectivity and the schematic

B. A minimum geometric spacing violation against a design-rule deck

C. A negative setup-slack result in a required timing scenario

D. An offset between patterned wafer layers measured as overlay

3. A colleague needs the term for this meaning: Process design kit: process-specific models, rules, and design resources. Which term fits?

A. PDK

B. engineering change order

C. engineering lot

D. sign-off

D Choose the response that fits / continued

4. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions." Ken: [missing response] Sofia: "Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled." Which response belongs in the gap?

A. Correct. A wafer start is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.

B. A waiver would need the applicable technical assessment and authorization. We cannot assume it is available or accepted simply because the earlier date is commercially attractive.

C. Report each DRC finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.

D. Keep the PDK revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.

Answers and explanations, page 159

One useful distinction

October 8 is a requested tape-out date, not an accepted slot or a finished-sample date. The open DRC and LVS items remain blockers to the stated sign-off package. The case does not provide an approved waiver or permission to bypass checks.

E Conversation

A possible slot is not tape-out readiness

A possible slot is not tape-out readiness

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

delivery commitment / Data acceptance / LVS / wafer start / tape-out / PDK / sign-off / waiver / capacity reservation / DRC

01 **Sofia:** Can we move tape-out from October fifteenth to October eighth? The foundry mentioned an earlier slot, and the customer is asking whether that means samples can arrive sooner.

02 **Ken:** Separate the [1] _____ request from the later milestones. An earlier data handoff may matter to the plan, but it does not itself establish a wafer start or sample-delivery date.

03 **Sofia:** We still have four design-rule findings and one layout-versus-schematic mismatch. I want to know whether those can remain open while we secure the opportunity.

04 **Ken:** The [2] _____ package is not complete. No waiver is approved, so we cannot describe the design as ready by treating the open checks as an informal schedule exception.

05 **Sofia:** The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions.

06 **Ken:** Keep the [3] _____ revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.

E Conversation / continued

07 **Sofia:** Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled.

08 **Ken:** Report each [4] _____ finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.

09 **Sofia:** The connectivity mismatch is a separate issue. I do not want the customer to hear that all five items are the same kind of geometric spacing check.

10 **Ken:** Correct. [5] _____ addresses layout versus schematic consistency. Keep that distinction clear when assigning ownership and explaining why the complete sign-off package is not yet available.

Conversation continues on the next page.

A possible slot is not tape-out readiness

CONVERSATION / CONTINUED

11 **Sofia:** On capacity, the foundry said a slot might become available. We have no written confirmation that it is allocated to this design.

12 **Ken:** Then there is no confirmed [6] _____. We can discuss feasibility, but a mentioned opportunity must not be reported as an accepted manufacturing commitment.

13 **Sofia:** Who confirms that the foundry has accepted the data package? I do not want a file-transfer receipt reported as acceptance of the design handoff.

14 **Ken:** Exactly. [7] _____ is a separate handoff status. Record what the foundry needs and who confirms it, rather than assume a sent dataset has cleared the receiving review.

15 **Sofia:** Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it.

16 **Ken:** A [8] _____ would need the applicable technical assessment and authorization. We cannot assume it is available or accepted simply because the earlier date is commercially attractive.

17 **Sofia:** I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff.

18 **Ken:** Correct. A [9] _____ is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.

A possible slot is not tape-out readiness / continued

19 **Sofia:** The next update will separate check closure from slot confirmation. It will identify owners and show what remains conditional rather than announce the earlier date as secured.

20 **Ken:** That protects the [10] _____. We can pursue the opportunity actively while making clear which evidence, acceptance, and manufacturing decisions still stand between a request and a confirmed outcome.

CHECK THE GAPS

1. tape-out / **2.** sign-off / **3.** PDK / **4.** DRC / **5.** LVS / **6.** capacity reservation / **7.** Data acceptance / **8.** waiver / **9.** wafer start / **10.** delivery commitment

Why these answers fit, page 161

F Check accuracy and scope

A possible slot is not tape-out readiness

Return to the case

Customer design lead Sofia asks foundry liaison Ken to move a planned October 15 tape-out to October 8. Four design-rule-check findings and one layout-versus-schematic mismatch remain unresolved in the agreed fictional PDK revision P3. The foundry has mentioned a possible earlier slot but has not reserved or accepted it. No complete sign-off package is available, and no waiver is approved. Ken must communicate the blockers and conditional next steps without promising a wafer start, finished samples, or shipment.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why retain the PDK and rule-deck versions with results?

A. The latest available deck replaces the agreed deck without an impact review.
B. The check result depends on the process-specific definitions actually used.
C. A clean result from one revision automatically clears later rule changes.
D. The design filename alone identifies every rule and model used.

2. Which statement about tape-out is accurate?

A. It always occurs on the same day as customer shipment.
B. It guarantees that no respin can be needed.
C. It is a design-data handoff milestone, not proof of wafer start or sample availability.
D. It means every package has completed reliability qualification.

Answers and explanations, page 162

Listen for the distinction

Switch roles for turns 5-16. Expand PDK, DRC, and LVS using the vocabulary pages and preserve each open status.

Separate the milestones under date pressure

Tape-out, foundry data acceptance, wafer start, and finished samples are related but distinct milestones. State which date is requested and which prerequisites remain open. A possible capacity opportunity should trigger a feasibility discussion, not erase design checks or become a shipment promise.

F Say it / transfer to a new case

A requested handoff is not a shipment

A team requests tape-out on November 3. Two DRC items remain open. The foundry has mentioned a possible slot but has not reserved it. No waiver or sample-delivery date is approved.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

wafer start / capacity reservation / tape-out / delivery commitment

01 **Designer:** Keep November third attached to the requested design-data handoff: [1] _____.

02 **Liaison:** The possible slot is not yet an allocated [2] _____.

03 **Designer:** I will not tell the customer that fabrication begins then; we have no confirmed [3] _____.

04 **Liaison:** And discussing a slot does not establish a finished-sample [4] _____.

CHECK THE GAPS

1. tape-out / **2.** capacity reservation / **3.** wafer start / **4.** delivery commitment

Why these answers fit, page 162

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "Who confirms that the foundry has accepted the data package? I do not want a file-transfer receipt reported as acceptance of the design handoff." Ken: [missing response] Sofia: "Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it." Which response belongs in the gap?

A. Report each DRC finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.

B. The sign-off package is not complete. No waiver is approved, so we cannot describe the design as ready by treating the open checks as an informal schedule exception.

C. Exactly. Data acceptance is a separate handoff status. Record what the foundry needs and who confirms it, rather than assume a sent dataset has cleared the receiving review.

D. Separate the tape-out request from the later milestones. An earlier data handoff may matter to the plan, but it does not itself establish a wafer start or sample-delivery date.

Answers and explanations, page 163

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Packaging, Test, and Reliability Qualification" and "Foundry, Tape-Out, PDK, and Capacity Communication". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Connecting a die face-down through conductive bumps or pillars. Which term fits?

A. QFN

B. thermal cycling

C. flip-chip attachment

D. test coverage

2. A colleague needs the term for this meaning: The implemented rules used by a physical-verification tool. Which term fits?

A. mask data preparation

B. wafer start

C. foundry

D. rule deck

3. Rebuild the exchange from "Same recipe, different means". Victor: "The observed difference needs investigation. Chamber history and maintenance condition may be relevant, but neither has been established as the cause from these numbers." Asha: [missing response] Victor: "I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?" Which response belongs in the gap?

A. The actual value has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.

B. Check calibration status through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.

C. That preserves the release status accurately. We can pursue the backup option while making its unresolved requirements visible to the people planning production.

D. Review chamber condition and the other appropriate records. Identical commanded settings do not guarantee identical physical conditions or measurement behavior.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff." Ken: [missing response] Sofia: "The next update will separate check closure from slot confirmation. It will identify owners and show what remains conditional rather than announce the earlier date as secured." Which response belongs in the gap?

A. Keep the PDK revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.

B. Correct. A wafer start is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.

C. Report each DRC finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.

D. A waiver would need the applicable technical assessment and authorization. We cannot assume it is available or accepted simply because the earlier date is commercially attractive.

Answers and explanations, page 173

Language for the next case

A failing wafer map may also tell a contact story

test bin

A test bin classifies the recorded test outcome; it does not by itself identify the physical cause of a failure.

Luca: Not yet. First separate the test bin from the cause. The bin tells us which test outcome was recorded, not why it happened.

site correlation

Site correlation examines whether the outcome follows a particular parallel test site, helping distinguish that pattern from the wafer's physical geography.

Luca: Exactly. Compare the site correlation with the die coordinates. Keep both views so we do not mistake the test arrangement for a wafer-location effect.

probe card

The probe card provides the electrical interface to wafer contacts; its condition and history can matter even when the test-program revision is unchanged.

Luca: No. The probe card, tester configuration, connections, and conditions still need review. A matching program name does not establish identical test behavior.

contact resistance

Contact resistance concerns the electrical resistance at the probing connection; relevant measurements can help investigate a contact hypothesis without proving it from a label.

Luca: I will retrieve the contact resistance records and the card history. At present, contact is a hypothesis, not a confirmed cause.

retest

Retest repeats testing under the applicable authorized conditions; it must preserve the first-pass record and identify the basis for interpreting a changed result.

Luca: Agreed. Any retest needs the authorized plan, traceable conditions, and a separate result. Repeating until something passes is not a justified investigation.

denominator

The denominator identifies the population used in a yield calculation; first-pass and selected-retest populations must not be silently interchanged.

Luca: Yes. Keep the denominator and population explicit, especially if only selected die are retested.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A failing wafer map may also tell a contact story

A wafer-sort run shows repeated failures at one probe site. The test program is unchanged, but contact-resistance information and probe-card history have not yet been reviewed. The test and yield engineers investigate. No cause, retest disposition, or shipment decision is established.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

test bin / contact resistance / probe card / retest / denominator / site correlation

01 **Nari:** The wafer map has a repeated pattern at one probe site. Can we tell the fab that this is a process defect?

02 **Luca:** Not yet. First separate the [1] _____ from the cause. The bin tells us which test outcome was recorded, not why it happened.

03 **Nari:** The pattern seems to follow the same site in successive touchdowns. That could be different from a defect following a physical wafer region.

04 **Luca:** Exactly. Compare the [2] _____ with the die coordinates. Keep both views so we do not mistake the test arrangement for a wafer-location effect.

05 **Nari:** The program revision is unchanged. Does that rule out the test system as a contributor?

06 **Luca:** No. The [3] _____, tester configuration, connections, and conditions still need review. A matching program name does not establish identical test behavior.

07 **Nari:** The operations note mentions intermittent contact. I would like to see the actual measurements rather than use that phrase as the explanation.

08 **Luca:** I will retrieve the [4] _____ records and the card history. At present, contact is a hypothesis, not a confirmed cause.

09 **Nari:** Please preserve the first-pass results too. If the team reruns selected die later, I do not want the original failures to vanish from the dataset.

10 **Luca:** Agreed. Any [5] _____ needs the authorized plan, traceable conditions, and a separate result. Repeating until something passes is not a justified investigation.

Conversation continues on the next page.

A failing wafer map may also tell a contact story

CONVERSATION / CONTINUED

11 **Nari:** Would a higher pass count after an intervention establish that every first-pass failure was only a contact problem?

12 **Luca:** No. We would need to compare the relevant evidence and understand the intervention. A changed result is useful, but it does not assign one cause to every failing die.

13 **Nari:** We should show first-pass yield and any later disposition separately. Otherwise the apparent improvement could simply reflect a different reporting rule.

14 **Luca:** Yes. Keep the [6] _____ and population explicit, especially if only selected die are retested.

15 **Nari:** I can compare the pattern across coordinates, sites, and lots. Is there anything you need from yield analysis before the equipment review?

16 **Luca:** The exact failing test identifiers and map references. That will let the equipment team connect its records to the same observations instead of a screenshot without context.

17 **Nari:** I will send those and keep the lot status unchanged. The customer update can say a site-associated pattern is under investigation.

18 **Luca:** Good. Do not say the fab caused it or the test system cleared it while those conclusions remain open.

19 **Nari:** Once the review establishes the next authorized work, we can report what was checked, what changed, and which conclusions the evidence supports.

20 **Luca:** Exactly. A useful wafer map leads to a focused investigation, not an immediate choice between blaming the process and discarding the test result.

CHECK THE GAPS

1. test bin / **2.** site correlation / **3.** probe card / **4.** contact resistance / **5.** retest / **6.** denominator

Why these answers fit, page 176

A selected retest is reported as the whole wafer yield

Only 20 initially failing die are retested. Fifteen pass on that retest. The complete first-pass wafer results remain available.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

population / results / yield / plan

01 **Yield engineer:** Fifteen of twenty describes the selected retest [1] _____.

02 **Test engineer:** Preserve the original first-pass [2] _____.

03 **Yield engineer:** Do not label seventy-five percent as the whole-wafer [3] _____.

04 **Test engineer:** Interpret any changed result under the authorized retest [4] _____.

CHECK THE GAPS

1. population / **2.** results / **3.** yield / **4.** plan

Why these answers fit, page 176

Keep the distinction audible

Discuss wafer-test evidence while distinguishing a test result, a contact hypothesis, and approved retest.

Language for the next case

Nominal timing passes, but signoff still has an open corner

setup slack

Setup slack compares the required and actual arrival timing for the setup check; negative slack means the stated constraint is not met in that scenario.

Rhea: The required slow-corner scenario has negative setup slack of zero point zero eight nanoseconds. Required hold analysis is also incomplete.

timing exception

A timing exception changes how specified paths are analyzed and requires a justified functional basis; it is not a way to hide an inconvenient violation.

Rhea: Only with a valid, reviewed timing exception. The fact that a path is inconvenient or infrequent does not prove it cannot be exercised under the relevant mode.

parasitic extraction

Parasitic extraction provides modeled interconnect effects such as resistance and capacitance; its configuration must match the design being analyzed.

Rhea: Confirm the parasitic extraction and other inputs before relying on the result. A precise slack number is not meaningful if the input configuration is inconsistent.

PVT corner

A PVT corner specifies process, voltage, and temperature assumptions for analysis; a pass at one corner does not automatically establish a pass at others.

Rhea: Yes. A PVT corner defines process, voltage, and temperature assumptions. Those conditions are part of the result, not a footnote we can drop.

hold analysis

Hold analysis checks minimum-delay timing requirements and is distinct from setup analysis; a setup improvement does not automatically establish hold correctness.

Rhea: It can. The hold analysis must be assessed on its own required basis. We should not assume one favorable change resolves every timing concern.

Timing closure

Timing closure is the process of meeting the defined timing requirements across the required analysis scope, not merely obtaining one favorable nominal report.

Rhea: Yes. It also needs the open checks and reviewed exceptions. Timing closure is against the required scope, not whichever report happens to be most favorable.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

Nominal timing passes, but signoff still has an open corner

A digital design passes the reviewed nominal setup analysis. One required slow-corner scenario reports setup slack of -0.08 ns, and required hold analysis is incomplete. A schedule note says timing clean. The timing lead and design engineer review the claim; no exception or signoff decision has been approved.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

PVT corner / parasitic extraction / hold analysis / timing exception / Timing closure / setup slack

01 **Jon:** Nominal timing passes, so I marked the design timing clean. You flagged the note. Which part of the signoff package is still open?

02 **Rhea:** The required slow-corner scenario has negative [1] _____ of zero point zero eight nanoseconds. Required hold analysis is also incomplete.

03 **Jon:** So the nominal result is useful, but it does not cover all the required operating assumptions. We need to show the failing scenario separately.

04 **Rhea:** Yes. A [2] _____ defines process, voltage, and temperature assumptions. Those conditions are part of the result, not a footnote we can drop.

05 **Jon:** The path is only used in a special mode. Could I declare it a false path to remove the violation from the report?

06 **Rhea:** Only with a valid, reviewed [3] _____. The fact that a path is inconvenient or infrequent does not prove it cannot be exercised under the relevant mode.

Conversation / continued

07 **Jon:** I will check the mode and constraints with verification. A false-path claim needs support from actual functional behavior, not the launch date.

08 **Rhea:** Exactly. Keep the endpoint, mode, and constraint reference with the question. We need to know whether the analysis is appropriate before deciding what design change is needed.

09 **Jon:** If we improve this setup path, can the same change affect a hold check elsewhere?

10 **Rhea:** It can. The [4] _____ must be assessed on its own required basis. We should not assume one favorable change resolves every timing concern.

Conversation continues on the next page.

Nominal timing passes, but signoff still has an open corner

CONVERSATION / CONTINUED

11 **Jon:** The latest report uses an updated netlist, but I am not sure the extracted interconnect data belongs to that revision.

12 **Rhea:** Confirm the [5] _____ and other inputs before relying on the result. A precise slack number is not meaningful if the input configuration is inconsistent.

13 **Jon:** Then the next update needs the exact netlist, constraints, libraries, extraction, modes, and corners. That is more useful than another screenshot of a green summary.

14 **Rhea:** Yes. It also needs the open checks and reviewed exceptions. [6] _____ is against the required scope, not whichever report happens to be most favorable.

15 **Jon:** I will remove timing clean and report nominal setup passed, one required setup violation open, and hold checks incomplete.

16 **Rhea:** That preserves the completed work without implying signoff. We can then assign the constraint review and the technical investigation to the right owners.

17 **Jon:** Should I promise a frequency reduction as the solution? It would change the customer requirement, so I think that belongs in a separate trade-off discussion.

18 **Rhea:** Correct. Do not quietly change the requirement to make the report pass. Any proposed specification change needs its own assessment and authorization.

19 **Jon:** I will bring the reviewed path evidence to the next meeting. We will distinguish a real design issue from a justified constraint correction rather than assume either one.

20 **Rhea:** Good. The final decision needs consistent inputs, completed required checks, and a supported treatment of exceptions, not a nominal pass expanded into a blanket claim.

CHECK THE GAPS

1. setup slack / **2.** PVT corner / **3.** timing exception / **4.** hold analysis / **5.** parasitic extraction / **6.** Timing closure

Why these answers fit, page 178

A negative slack is hidden by an unsupported exception

A required timing scenario has negative slack. An engineer proposes excluding the path without evidence that the exception is functionally valid.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

mode / revision / basis / open

01 **Reviewer:** Keep the violation visible until the exception has a justified [1] _____.

02 **Engineer:** Check the path against the relevant operating [2] _____.

03 **Reviewer:** Preserve the actual analysis inputs and configuration [3] _____.

04 **Engineer:** Do not call timing closed while required checks remain [4] _____.

CHECK THE GAPS

1. basis / **2.** mode / **3.** revision / **4.** open

Why these answers fit, page 178

Keep the distinction audible

Explain a timing violation and challenge an unjustified exception without confusing analysis scopes.

Language for the next case

A resealed bag does not explain the exposure history

moisture sensitivity level

The moisture sensitivity level identifies a package's classification for moisture-related handling; it is not evidence that this reel's actual exposure stayed within its requirements.

Caleb: We need the history. The moisture sensitivity level on the label does not tell us how this particular reel was handled after opening.

floor life

Floor life concerns the permitted exposure under specified conditions before the relevant assembly process; a new bag seal alone does not establish a fresh exposure allowance.

Caleb: Resealing alone does not establish a reset of floor life. We must follow the applicable product and handling requirements, not invent a new allowance.

cumulative exposure

Cumulative exposure concerns the relevant accumulated time and conditions after opening; an undocumented interval cannot be treated as zero simply because the reel was resealed.

Caleb: Please distinguish a confirmed record from an estimate. We need the cumulative exposure and storage conditions, including any interval we cannot account for.

humidity indicator card

A humidity indicator card provides an indication relevant to humidity within the package under its specified interpretation; it does not reconstruct the complete historical exposure record.

Caleb: No. The humidity indicator card is one part of the applicable assessment. It is not a complete time history of every earlier handling event.

Shelf life

Shelf life concerns storage over a defined period under stated conditions; it is a different question from post-opening floor life before assembly.

Caleb: Correct. Shelf life and post-opening exposure are different questions. Keep the packaging and storage conditions attached to either statement.

disposition

Disposition is the authorized decision about the material's next route after reviewing the relevant evidence and requirements; resealing does not supply that decision.

Caleb: Yes. The disposition must be recorded. A neat bag and a current label are not the same as an authorized material decision.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A resealed bag does not explain the exposure history

A reel of moisture-sensitive packaged devices arrives at assembly in a resealed bag. The original opening time and cumulative exposure record are missing. The package label identifies the part and moisture sensitivity level, but no approved disposition is recorded. A materials coordinator asks a packaging engineer how to describe the status.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

Shelf life / floor life / humidity indicator card / moisture sensitivity level / cumulative exposure / disposition

01 **Sora:** The reel is back in a sealed bag. Can I mark it ready for assembly, or do you still need the missing opening-time record?

02 **Caleb:** We need the history. The [1] _____ on the label does not tell us how this particular reel was handled after opening.

03 **Sora:** The bag has a recent resealing date. Someone suggested using that as a fresh start for the clock.

04 **Caleb:** Resealing alone does not establish a reset of [2] _____. We must follow the applicable product and handling requirements, not invent a new allowance.

05 **Sora:** I will look for the material movement records and ask the previous shift when the bag was first opened.

06 **Caleb:** Please distinguish a confirmed record from an estimate. We need the [3] _____ and storage conditions, including any interval we cannot account for.

Conversation / continued

07 **Sora:** There is a humidity indicator card in the bag. Does its current appearance prove that the reel was always stored correctly?

08 **Caleb:** No. The [4] _____ is one part of the applicable assessment. It is not a complete time history of every earlier handling event.

09 **Sora:** The supplier also gives shelf-life information. I should not use the remaining months there as an answer to the missing floor-life record.

10 **Caleb:** Correct. [5] _____ and post-opening exposure are different questions. Keep the packaging and storage conditions attached to either statement.

Conversation continues on the next page.

A resealed bag does not explain the exposure history

CONVERSATION / CONTINUED

11 **Sora:** Could the team simply choose a baking cycle from a similar component? I do not want a language misunderstanding to become an improvised process instruction.

12 **Caleb:** No. Any conditioning must follow the applicable approved requirements for the actual part, package, and materials. This conversation does not prescribe a temperature or duration.

13 **Sora:** Then I will keep the reel identified and unavailable for routine issue while the responsible team reviews the missing history under our procedure.

14 **Caleb:** Yes. The [6] _____ must be recorded. A neat bag and a current label are not the same as an authorized material decision.

15 **Sora:** I can give production an update now: exact reel identified, exposure history incomplete, review requested. I should not call the devices damaged either.

16 **Caleb:** That is accurate. Missing evidence means the status is unresolved; it does not establish a physical failure in every device.

17 **Sora:** If we find the original records, I will send them with their source and timestamps rather than replacing the unknown interval with a recollection.

18 **Caleb:** Good. Preserve the difference between the recovered record and anyone's estimate so the reviewer can assess the actual basis.

19 **Sora:** I will also ask the team to improve the handover field for opening time. We should not face the same missing-history question on the next reel.

20 **Caleb:** Agreed. Resolve this reel through the approved process, then address the record gap without claiming that a better form retrospectively proves its handling was acceptable.

CHECK THE GAPS

1. moisture sensitivity level / **2.** floor life / **3.** cumulative exposure / **4.** humidity indicator card / **5.** Shelf life / **6.** disposition

Why these answers fit, page 180

Shelf life is used to dismiss an unknown exposure interval

A device reel is within its stated storage shelf life, but its post-opening exposure history is incomplete.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

requirements / issue / history / record

01 **Coordinator:** Storage shelf life does not resolve the missing exposure [1] _____.

02 **Engineer:** Identify the actual part, package, and applicable handling [2] _____.

03 **Coordinator:** Keep unknown time separate from a confirmed [3] _____.

04 **Engineer:** Await the authorized material decision before routine [4] _____.

CHECK THE GAPS

1. history / **2.** requirements / **3.** record / **4.** issue

Why these answers fit, page 180

Keep the distinction audible

Clarify moisture-handling records and distinguish shelf life, floor life, and disposition.

01 / Answers and explanations

A / Read the situation

A1. Where is L24 now?

D. In wafer fabrication at an M2-related operation

The briefing places L24 in wafer fabrication, not the packaging stage Eric inferred.

Return to the exercise, page 4

A2. What does BEOL mean in this case?

A. Back end of line, forming on-wafer interconnects

The integration engineer uses BEOL for on-wafer interconnect fabrication.

Return to the exercise, page 4

A3. What prevents the next operation?

C. A pending inspection review and lot hold

The lot remains on hold until the stated inspection review and authorized disposition.

Return to the exercise, page 4

D / Response choices

D1. Which customer update is accurate?

A. L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.

M2 identifies the current fabrication work, not completion of the stack or assembly. The hold still prevents an unapproved next move.

Return to the exercise, page 8

D2. Why expand BEOL for the coordinator?

D. Different teams can use back-end for different parts of the manufacturing chain.

Expanding the term prevents a cross-team ambiguity that already caused an inaccurate update.

Return to the exercise, page 8

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Processes that form devices and interconnects on wafers. Which term fits?

D. wafer fabrication

A Not here. disposition: An authorized decision about the next treatment or status of material. The usual combination is "obtain lot disposition".

B Not here. lot: A tracked group of wafers or units processed under defined identity. The usual combination is "identify the lot".

C Not here. BEOL: Back end of line, forming the on-wafer interconnect structure. The usual combination is "track BEOL progress".

D Correct. wafer fabrication: Processes that form devices and interconnects on wafers. The usual combination is "track wafer fabrication".

Return to the exercise, page 8

D4. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I will name M2 in the update. Does that identify the current layer, rather than mean the full interconnect stack is finished?" Min: [missing response] Eric: "Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab." Which response belongs in the gap?

D. Correct. M2 is the relevant metal layer. A layer name locates the work, but it does not by itself tell you that inspection, disposition, or later stages have finished.

A Not here. This line belongs to turn 8, after Eric says: "Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Eric says: "That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Eric says: "I wrote that L24 is in packaging because the status says BEOL. The customer wants to know whether that means finished parts will be ready soon." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Min's response in turn 6. It connects Eric's preceding point with Eric's follow-up. The metal layer identifies the specific interconnect level involved in the current operation.

Return to the exercise, page 8

E / Conversation

1. BEOL

Turn 2. BEOL refers to the on-wafer interconnect stage in this case, not to completed package assembly.

01 / Answers and explanations / continued

2. wafer fabrication

Turn 4. Wafer fabrication locates the current work on the wafer before the package assembly stage described in the mistaken update.

3. metal layer

Turn 6. The metal layer identifies the specific interconnect level involved in the current operation.

4. FEOL

Turn 8. FEOL principally forms active devices, distinct from the later on-wafer interconnect work now involving L24.

5. lot hold

Turn 10. A lot hold restricts further movement until the required review and authorized disposition occur.

6. disposition

Turn 12. Disposition is the authorized decision about how the lot may proceed after review, not an assumption based on its position in the flow.

7. traveler

Turn 14. The traveler connects the material with its operation history and required next steps, supporting traceable status communication.

8. Wafer sort

Turn 16. Wafer sort is electrical testing of die on the wafer and is distinct from package assembly or shipment.

9. package assembly

Turn 18. Package assembly is the later integration of die into packages, which has not started for L24.

10. process flow

Turn 20. The process flow connects the current stage and authorized next steps without collapsing fabrication, testing, and assembly into one label.

Return to the conversation, page 10

01 / Answers and explanations / continued

F / Accuracy and scope

F1. What is the appropriate next-step wording?

B. The lot awaits inspection review and authorized disposition under its route.

The stated hold and route control what may happen next, not a generic assumption about progress.

Return to the exercise, page 13

F2. Which distinction is correct?

A. Wafer sort tests die on the wafer; package assembly integrates die into packages.

Wafer sort is electrical testing, dicing is physical separation, FEOL principally forms devices, and BEOL forms on-wafer interconnects.

Return to the exercise, page 13

F / Guided speaking

1. BEOL

Coordinator: I will correct the update: this on-wafer interconnect work is still BEOL.

Turn 1. BEOL names the on-wafer interconnect stage, not package assembly.

2. lot hold

Engineer: Keep the restriction visible too; the record still shows a lot hold.

Turn 2. A lot hold restricts movement while the required review and decision remain open.

3. package assembly

Coordinator: Then I should not tell the customer we have started package assembly.

Turn 3. Package assembly is distinct from the current on-wafer work and has not begun here.

4. disposition

Engineer: Correct. Ask the review owner for the authorized disposition before discussing the next move.

Turn 4. Disposition is the authorized material decision, not a date inferred from the process stage.

Return to the conversation, page 14

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I will confirm the owner and preserve the lot reference. We should also keep the record connected to the individual wafers where the inspection information requires that detail." Min: [missing response] Eric: "What later stages should I distinguish in the message? I need to explain why an M2 status does not yet give the customer a finished-product date." Which response belongs in the gap?

A. Keep the traveler and relevant wafer IDs aligned with the update. Otherwise two teams may discuss different material while assuming they are referring to the same issue.

A Correct. This is Min's response in turn 14. It connects Eric's preceding point with Eric's follow-up. The traveler connects the material with its operation history and required next steps, supporting traceable status communication.

B Not here. This line belongs to turn 4, after Eric says: "That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Eric says: "Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Eric says: "Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 14

02 / Answers and explanations

A / Read the situation

A1. Which layer and stage do the records identify?

D. M2 after etch

The recovered records specifically locate the two means on M2 after etch.

Return to the exercise, page 16

A2. What is the reported numerical difference?

C. An increase of 3 nm

Subtracting the first mean, 86 nm, from the second, 89 nm, gives a 3 nm increase.

Return to the exercise, page 16

A3. Why is a direct process attribution premature?

C. The measurement recipes and sampling sites differ.

Changed measurement and sampling conditions prevent the difference from isolating a process cause.

Return to the exercise, page 16

D / Response choices

D1. Which report sentence is justified?

C. The means differ by 3 nm, but recipe and site changes limit comparability.

The sentence reports the arithmetic while preserving the unresolved measurement basis.

Return to the exercise, page 20

D2. Which comparison best helps investigate the apparent change?

C. Comparable measurements on identified sites using an agreed recipe basis

An agreed measurement basis addresses the changed recipes and sampling. More data alone does not remove those differences, and different stages answer different questions.

Return to the exercise, page 20

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The delivered exposure energy per unit area. Which term fits?

D. exposure dose

A Not here. line-edge roughness: Variation of a patterned line edge from its intended shape. The usual combination is "assess line-edge roughness".

B Not here. sampling site: A specified location selected for measurement. The usual combination is "align sampling sites".

C Not here. depth of focus: A usable focus range under a defined imaging criterion. The usual combination is "assess depth of focus".

D Correct. exposure dose: The delivered exposure energy per unit area. The usual combination is "record exposure dose".

Return to the exercise, page 20

D4. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting." Theo: [missing response] Lina: "I will keep the three-nanometer difference and remove the exposure-drift conclusion. Can we show the changed recipe and sites beside the two means?" Which response belongs in the gap?

A. The sampling site selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.

A Correct. This is Theo's response in turn 6. It connects Lina's preceding point with Lina's follow-up. A sampling site is a selected measurement location; different site selections can affect the reported average.

B Not here. This line belongs to turn 20, after Lina says: "The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Lina says: "Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 14, after Lina says: "Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 20

02 / Answers and explanations / continued

E / Conversation

1. post-etch

Turn 2. Post-etch locates the measurement after pattern transfer, which matters when interpreting possible causes.

2. measurement recipe

Turn 4. The measurement recipe defines settings and analysis that can affect the reported CD and differs between these datasets.

3. sampling site

Turn 6. A sampling site is a selected measurement location; different site selections can affect the reported average.

4. critical dimension

Turn 8. Critical dimension is the relevant feature size, but its reported change does not alone establish the physical cause.

5. Etch bias

Turn 10. Etch bias concerns the dimensional change associated with pattern transfer and is one reason post-etch CD does not isolate exposure.

6. Overlay

Turn 12. Overlay describes alignment between layers, whereas CD describes a feature dimension.

7. remeasurement

Turn 14. Remeasurement on an agreed basis can help determine whether the apparent difference persists under comparable conditions.

8. measurement bias

Turn 16. Measurement bias is a systematic difference in measurement relative to a reference and is distinct from an assumed process shift.

9. reticle

Turn 18. The reticle carries the lithographic pattern; identifying its revision provides context without proving it caused the observation.

02 / Answers and explanations / continued

10. pattern fidelity

Turn 20. Pattern fidelity concerns how well the produced pattern matches its intent; assessing it requires more than an unexplained mean difference.

Return to the conversation, page 22

F / Accuracy and scope

F1. Why is post-etch important?

B. The result follows pattern transfer and does not isolate lithography alone.

The measurement stage affects interpretation because more than the exposure step contributes to the resulting feature.

Return to the exercise, page 26

F2. Which distinction is accurate?

A. CD concerns feature size; overlay concerns layer alignment.

CD and overlay separate size from alignment. Pitch is repeat distance, roughness is edge variation, dose is exposure energy per area, and etch bias concerns dimensional transfer.

Return to the exercise, page 26

F / Guided speaking

1. critical dimension

Engineer: Check that both reports refer to the same critical dimension.

Turn 1. Critical dimension refers to feature size, not the settings or locations used to measure it.

2. measurement recipe

Reviewer: Send the settings and analysis from each measurement recipe.

Turn 2. The recipe defines the settings and analysis that differ between the datasets.

3. sampling sites

Engineer: I will attach a wafer map of both sets of sampling sites.

Turn 3. Sampling sites identify the measurement locations, which also differed in this case.

4. overlay

Reviewer: Good. An alignment claim would need separate overlay evidence.

Turn 4. Overlay addresses alignment rather than the reported feature-size difference of four nanometers.

Return to the conversation, page 27

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis." Theo: [missing response] Lina: "If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy." Which response belongs in the gap?

D. Request comparable remeasurement through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.

A Not here. This line belongs to turn 10, after Lina says: "Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 6, after Lina says: "Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Lina says: "The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Theo's response in turn 14. It connects Lina's preceding point with Lina's follow-up. Remeasurement on an agreed basis can help determine whether the apparent difference persists under comparable conditions.

Return to the exercise, page 27

03 / Answers and explanations

A / Read the situation

A1. What improved in the supplied trial?

B. The stated nonuniformity metric fell from 5% to 3% at the nominal setting.

The only reported improvement concerns the unchanged metric at one tested condition.

Return to the exercise, page 30

A2. What is the absolute change in that percentage metric?

A. A decrease of 2 percentage points

Five percent minus three percent equals two percentage points on the stated metric.

Return to the exercise, page 30

A3. Which conclusion is premature?

C. The complete process window is validated.

The broader range and downstream behavior were not assessed, so full-window validation is unsupported.

Return to the exercise, page 30

D / Response choices

D1. Which summary preserves the evidence boundary?

B. The nominal-setting metric improved; the range and downstream effects remain untested.

The observed improvement concerns one response at one setting. Untested settings, other film properties, integration, and release remain separate questions.

Return to the exercise, page 35

D2. What is the relative reduction from 5% to 3%?

C. 40%

The decrease of two divided by the original five equals forty percent.

Return to the exercise, page 35

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Physical vapor deposition, forming a film from material vaporized by methods such as sputtering or evaporation. Which term fits?

A. PVD

A Correct. PVD: Physical vapor deposition, forming a film from material vaporized by methods such as sputtering or evaporation. The usual combination is "review PVD performance".

B Not here. CMP: Chemical mechanical planarization, combining chemical and mechanical surface removal. The usual combination is "evaluate CMP compatibility".

C Not here. corner condition: A selected limiting combination in an evaluation range. The usual combination is "evaluate corner conditions".

D Not here. film stress: Internal stress within a deposited film. The usual combination is "measure film stress".

Return to the exercise, page 35

D4. Rebuild the exchange from "A better center point is not a wider window". Yuki: "The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent." Sam: [missing response] Yuki: "I will mark the low and high settings untested. What evidence does the approved plan require before we can make a claim about that range?" Which response belongs in the gap?

C. Keep that arithmetic tied to the nominal setting. Neither way of expressing the improvement extends the evidence to settings that have not been run.

A Not here. This line belongs to turn 4, after Yuki says: "Both values use the same measurement definition, and lower is better for this metric. I want to report that improvement clearly without losing the limitation." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 20, after Yuki says: "I will revise the summary to report the center-point improvement, the unchanged metric basis, and the untested range. I will list the etch and planarization reviews as pending." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Sam's response in turn 6. It connects Yuki's preceding point with Yuki's follow-up. The nominal setting is the only process condition tested, so the measured improvement must remain limited to it.

D Not here. This line belongs to turn 18, after Yuki says: "Please keep trial revision on the summary. Planning is asking about routine use, and I do not want this result mistaken for a production release." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 35

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03 / Answers and explanations / continued

E / Conversation

1. process window

Turn 2. A process window is a range satisfying defined requirements, not a single successful nominal setting.

2. thickness nonuniformity

Turn 4. Thickness nonuniformity is the specific response that improved; it does not stand for all film or integration properties.

3. nominal setting

Turn 6. The nominal setting is the only process condition tested, so the measured improvement must remain limited to it.

4. corner condition

Turn 8. A corner condition is a defined limiting setting or combination that requires its own evidence in the evaluation plan.

5. integration trade-off

Turn 10. An integration trade-off concerns benefits and consequences across related steps that a single improved deposition metric cannot settle.

6. etch selectivity

Turn 12. Etch selectivity compares material removal rates and may be affected by a film change; no result is supplied here.

7. CMP

Turn 14. CMP is chemical mechanical planarization, a downstream process whose compatibility with the changed film remains unassessed.

8. process interaction

Turn 16. A process interaction is a relationship between factors or steps that can change the overall result and needs explicit assessment.

9. recipe release

Turn 18. Recipe release is the controlled authorization for defined use, separate from a promising experimental result.

03 / Answers and explanations / continued

10. deposition

Turn 20. Deposition is the film-forming step actually assessed; the statement should not imply the entire integrated flow was validated.

Return to the conversation, page 37

F / Accuracy and scope

F1. Why review downstream etch and CMP?

B. A film change may alter later process behavior.

A changed film can interact differently with later processes. A shared recipe or one improved response does not establish selectivity, polishing behavior, or all material properties.

Return to the exercise, page 41

F2. Which status should remain separate from trial success?

B. Recipe release

Recipe release is an authorized decision that has not occurred in this case.

Return to the exercise, page 41

F / Guided speaking

1. nominal setting

Engineer: The metric fell from eight to six percent, but only at the nominal setting.

Turn 1. The nominal setting is the single tested condition, not the full range.

2. process window

Reviewer: Keep the low and high settings open; we have not established the process window.

Turn 2. A process window requires evidence across the relevant range, which is missing here.

3. integration question

Engineer: I will also list the untested effect on the later etch as an integration question.

Turn 3. The film change may affect a later step, so downstream compatibility remains an integration question.

4. recipe release

Reviewer: And keep routine production use pending the authorized recipe release.

Turn 4. A favorable trial result does not itself authorize the recipe for production use.

Return to the conversation, page 42

03 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A better center point is not a wider window". Yuki: "The planarization team also needs the proposed film information. Their review has not started, so I should not imply that the next surface-finishing step is already covered." Sam: [missing response] Yuki: "We can identify the relevant responses in the evaluation plan without improvising operating settings during this review. The process owners should define the approved technical work." Which response belongs in the gap?

C. State that CMP compatibility remains open. The deposition result does not establish that planarization behavior, surface effects, or related requirements are unchanged.

A Not here. This line belongs to turn 6, after Yuki says: "The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Yuki says: "Both values use the same measurement definition, and lower is better for this metric. I want to report that improvement clearly without losing the limitation." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Sam's response in turn 14. It connects Yuki's preceding point with Yuki's follow-up. CMP is chemical mechanical planarization, a downstream process whose compatibility with the changed film remains unassessed.

D Not here. This line belongs to turn 18, after Yuki says: "Please keep trial revision on the summary. Planning is asking about routine use, and I do not want this result mistaken for a production release." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 42

04 / Answers and explanations

A / Read the situation

A1. What is the pilot yield?

B. 92% from 92 of 100 tested die

Ninety-two passing die divided by one hundred tested gives a 92% observed yield.

Return to the exercise, page 44

A2. What is the absolute difference between the observed yields?

C. 2 percentage points

The observed yields are 92% and 90%, a difference of two percentage points.

Return to the exercise, page 44

A3. Which factor limits the causal comparison?

C. The product mixes differ.

Different product mixes can affect yield, so the process change is not isolated by this comparison.

Return to the exercise, page 44

D / Response choices

D1. Which announcement is supported?

D. The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.

The counts support descriptive percentages, but the comparison does not isolate cause or establish stability.

Return to the exercise, page 48

D2. What is the approximate relative increase from 90% to 92%?

B. 2.22%

The two-point increase divided by the 90% starting value is approximately 2.22%.

Return to the exercise, page 48

04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The set of items included in a reported test result. Which term fits?

A. tested population

A Correct. tested population: The set of items included in a reported test result. The usual combination is "define the tested population".

B Not here. percentage point: A unit for the absolute difference between percentages. The usual combination is "report percentage-point change".

C Not here. subgroup: A defined set of observations grouped for process analysis. The usual combination is "define rational subgroups".

D Not here. control limit: A monitoring boundary derived under a stated process model or method. The usual combination is "distinguish control limits".

Return to the exercise, page 48

D4. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points." Ravi: [missing response] Elena: "I want the team to see the pilot as useful progress without overstating what caused the result. The change happened before the pilot, but timing is not enough." Which response belongs in the gap?

D. Yes, use percentage point for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.

A Not here. This line belongs to turn 10, after Elena says: "Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 18, after Elena says: "We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Elena says: "The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ravi's response in turn 6. It connects Elena's preceding point with Elena's follow-up. Percentage point measures the absolute difference between the two yield percentages, here 92 minus 90.

Return to the exercise, page 48

04 / Answers and explanations / continued

E / Conversation

1. pass count

Turn 2. The pass count identifies the actual successes behind the percentage, preserving the scale of the evidence.

2. product mix

Turn 4. Product mix differs between the lots and may affect observed yield independently of the process change.

3. percentage point

Turn 6. Percentage point measures the absolute difference between the two yield percentages, here 92 minus 90.

4. causal attribution

Turn 8. Causal attribution assigns the observed yield difference to the process change, which this mixed comparison does not isolate.

5. stratification

Turn 10. Stratification separates the data into comparable product groups to examine the mix effect more directly.

6. process stability

Turn 12. Process stability concerns statistical behavior over time under a defined monitoring method, not two favorable aggregate yields.

7. specification limit

Turn 14. A specification limit defines acceptance requirements and is distinct from a statistically based process-monitoring boundary.

8. control signal

Turn 16. A control signal triggers investigation under the monitoring method; it does not provide its own causal diagnosis.

9. measurement system

Turn 18. The measurement system includes methods and conditions that affect whether the collected results can be compared consistently.

04 / Answers and explanations / continued

10. yield learning

Turn 20. Yield learning uses the observations to investigate losses and drivers without claiming an effect before the evidence supports it.

Return to the conversation, page 50

F / Accuracy and scope

F1. Which next analysis addresses the product-mix problem?

B. Separate comparable product groups and review consistent test bases.

Stratification examines comparable product groups. Equal sample sizes, pooled totals, or pass counts alone do not remove a difference in product composition.

Return to the exercise, page 53

F2. Which statement distinguishes control and specification limits?

A. Control limits monitor behavior; specification limits define acceptance boundaries.

Statistical stability and conformity are different questions. A stable process may miss specifications, and in-spec observations do not establish statistical control.

Return to the exercise, page 53

F / Guided speaking

1. denominator

Engineer: Put 48 passing die over the trial's 50-die denominator.

Turn 1. The tested total is the denominator; the 48 passing die form the numerator.

2. product mix

Reviewer: Then show the changed product mix beside the reference comparison, not in a separate appendix.

Turn 2. Product mix differs between the trial and reference and may affect the observed yields.

3. percentage points

Engineer: I will report the difference as six percentage points, not six relative percent.

Turn 3. Ninety-six percent minus ninety percent is six percentage points, not six relative percent.

04 / Answers and explanations / continued

4. causal attribution

Reviewer: Good. Leave the claim that the process caused it open until we have supported causal attribution.

Turn 4. Causal attribution needs evidence isolating the process effect, which the supplied comparison lacks.

Return to the conversation, page 54

F / Rebuild the exchange

1. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "We have acceptance criteria for individual test results. Some colleagues are treating those limits as though they were the same as the chart's monitoring boundaries." Ravi: [missing response] Elena: "If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it." Which response belongs in the gap?

D. A specification limit defines an acceptance boundary. A control limit has a different purpose, so passing the specification is not proof that the process is in statistical control.

A Not here. This line belongs to turn 12, after Elena says: "The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Elena says: "If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Elena says: "The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ravi's response in turn 14. It connects Elena's preceding point with Elena's follow-up. A specification limit defines acceptance requirements and is distinct from a statistically based process-monitoring boundary.

Return to the exercise, page 54

05 / Answers and explanations

A / Read the situation

A1. What does the inspection record contain?

A. 24 particle-like detections over 200 square centimeters

The record describes surface inspection detections and inspected area, not confirmed failure or added-defect counts.

Return to the exercise, page 58

A2. What does the map show?

D. An edge cluster

The supplied observation is a spatial edge cluster, without an established source.

Return to the exercise, page 58

A3. Why cannot the 24 detections be called 24 added defects?

C. No comparable pre-process scan exists, and review is incomplete.

Added defects require a valid before-and-after basis, and the detections are not all confirmed defects.

Return to the exercise, page 58

D / Response choices

D1. What is the observed detection density?

B. 0.12 detections per square centimeter

Twenty-four detections divided by two hundred square centimeters equals 0.12 on a surface-area basis.

Return to the exercise, page 62

D2. Which conclusion is unsupported?

D. C2 introduced all 24 detections.

Timing after C2 does not establish introduction by C2 without the needed comparison and investigation.

Return to the exercise, page 62

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The surface area included in an inspection result. Which term fits?

A. inspected area

A Correct. inspected area: The surface area included in an inspection result. The usual combination is "state the inspected area".

B Not here. airborne particle count: A count of particles in a defined air sample or volume. The usual combination is "review airborne particle counts".

C Not here. size threshold: The size boundary used for counting or classifying features. The usual combination is "state the size threshold".

D Not here. wafer map: A spatial representation of results on a wafer. The usual combination is "examine the wafer map".

Return to the exercise, page 62

D4. Rebuild the exchange from "Twenty-four detections are not a confirmed source". Hana: "The map clusters near the edge. I will send the wafer IDs and coordinates so the team can examine the pattern without guessing its source." Oscar: [missing response] Hana: "The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan." Which response belongs in the gap?

A. Call it an edge cluster. Describe where it appears and connect the map to the wafer IDs, rather than turn the pattern into an immediate diagnosis of the cleanroom.

A Correct. This is Oscar's response in turn 6. It connects Hana's preceding point with Hana's follow-up. An edge cluster describes the spatial concentration actually observed without claiming its origin has been established.

B Not here. This line belongs to turn 2, after Hana says: "The review note says cleanroom problem, but that label is much broader than the evidence. We have twenty-four particle-like detections on the inspected surfaces from P8." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Hana says: "The corrected note will give the count, area, edge pattern, interval, and incomplete classification. It will state that introduction by C2 and an airborne source are unconfirmed." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Hana says: "A matched earlier inspection would help distinguish what was already present from what appeared later. We would also need to check the inspection basis before comparing counts." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 62

05 / Answers and explanations / continued

E / Conversation

1. inspected area

Turn 2. The inspected area provides the denominator for the surface detection density rather than leaving the count without scale.

2. Defect review

Turn 4. Defect review determines how flagged features should be classified before they are treated as confirmed relevant defects.

3. edge cluster

Turn 6. An edge cluster describes the spatial concentration actually observed without claiming its origin has been established.

4. added defect

Turn 8. An added defect requires evidence that it appeared between comparable inspection points, which is missing here.

5. pre-process scan

Turn 10. A pre-process scan provides the earlier observation needed for a valid before-and-after comparison when methods are comparable.

6. airborne particle count

Turn 12. An airborne particle count concerns a defined air sample or volume, unlike the surface-area-based detections here.

7. inspection sensitivity

Turn 14. Inspection sensitivity affects which features are detected and must be considered when comparing inspection results.

8. containment scope

Turn 16. Containment scope identifies which material or operations are restricted while the investigation remains open.

9. exposure history

Turn 18. Exposure history connects the wafers to relevant handling and environments that may help evaluate possible contamination origins.

05 / Answers and explanations / continued

10. contamination source

Turn 20. The contamination source is the origin to be established by investigation, not assumed from the latest process step or vague label.

Return to the conversation, page 64

F / Accuracy and scope

F1. Which next record is useful for the investigation?

C. Traceable wafer, inspection-setting, timing, and exposure information

The traceable records preserve the actual material and measurement context. Unmatched layers, aggregate totals, and unsupported conversions cannot replace that evidence.

Return to the exercise, page 67

F2. Which distinction is correct?

A. Surface detection density and airborne particle counts use different measurement bases.

Surface and air results have different bases. Detection is not confirmed classification, timing is not proof of introduction, and a pattern alone does not identify a source.

Return to the exercise, page 67

F / Guided speaking

1. inspected area

Engineer: Keep 100 square centimeters beside the count as the inspected area.

Turn 1. Inspected area supplies the surface basis for the 12 detections, giving a density of 0.12.

2. defect review

Reviewer: And leave the twelve flagged features unclassified pending defect review.

Turn 2. Defect review determines how inspection detections should be classified; it remains incomplete.

3. pre-process scan

Engineer: I will request a comparable pre-process scan before reporting any process-added count.

Turn 3. A comparable pre-process scan is needed to assess what was introduced between inspection points.

05 / Answers and explanations / continued

4. contamination source

Reviewer: Thank you. Do not name the last tool as the contamination source without supporting investigation.

Turn 4. The source is an origin to investigate, not a conclusion established by the post-process count.

Return to the conversation, page 68

F / Rebuild the exchange

1. Rebuild the exchange from "Twenty-four detections are not a confirmed source". Hana: "I will include the inspection recipe and threshold with the maps. That will let reviewers see whether the method was capable of detecting the features being discussed." Oscar: [missing response] Hana: "P8 remains on hold under the local procedure. The investigation team needs to say which material and time interval are included, rather than leave the restriction implicit." Which response belongs in the gap?

A. Keep inspection sensitivity visible. A tool finding more features after a settings change does not by itself prove that the process produced more contamination.

A Correct. This is Oscar's response in turn 14. It connects Hana's preceding point with Hana's follow-up. Inspection sensitivity affects which features are detected and must be considered when comparing inspection results.

B Not here. This line belongs to turn 10, after Hana says: "A matched earlier inspection would help distinguish what was already present from what appeared later. We would also need to check the inspection basis before comparing counts." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Hana says: "The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Hana says: "That gives zero point one two detections per square centimeter. I should keep the word detections because the detailed classification has not been completed." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 68

06 / Answers and explanations

A / Read the situation

A1. How far apart are the sample means?

D. 4 nm

The difference between 102 nm and 98 nm is four nanometers.

Return to the exercise, page 70

A2. Do the means meet the stated matching criterion?

C. No; 4 nm exceeds the allowed 2 nm difference.

The matching criterion applies to the difference between means, not merely each mean's specification status.

Return to the exercise, page 70

A3. What does 90% describe here?

A. Availability for 900 of the defined 1,000 minutes

The supplied time ratio concerns equipment availability and establishes no product-yield or approval claim.

Return to the exercise, page 70

D / Response choices

D1. Which status report is accurate?

C. Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.

The specification and matching tests use different criteria, and individual results are not supplied.

Return to the exercise, page 74

D2. What does using the same recipe establish?

B. The identified recipe input is shared, not that output performance is interchangeable.

Shared commanded inputs do not establish identical physical conditions, equivalent output, or authorization to transfer the process.

Return to the exercise, page 74

06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A commanded target value for a controlled variable. Which term fits?

A. setpoint

A Correct. setpoint: A commanded target value for a controlled variable. The usual combination is "record the setpoint".

B Not here. scheduled downtime: Planned time when equipment is unavailable for the defined use. The usual combination is "plan scheduled downtime".

C Not here. chamber condition: The physical and operating state of a process chamber. The usual combination is "assess chamber condition".

D Not here. process transfer: Moving a defined process to another tool or location. The usual combination is "approve process transfer".

Return to the exercise, page 74

D4. Rebuild the exchange from "Same recipe, different means". Victor: "Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed." Asha: [missing response] Victor: "I will say both sample means meet that limit. We have not reviewed every individual measurement, so I cannot extend that claim to every wafer." Which response belongs in the gap?

D. The actual value has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.

A Not here. This line belongs to turn 14, after Victor says: "For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Victor says: "We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 12, after Victor says: "I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Asha's response in turn 6. It connects Victor's preceding point with Victor's follow-up. Actual value concerns the measured outcome; passing the mean specification does not remove the separate difference between tools.

Return to the exercise, page 74

E / Conversation

1. Tool matching

Turn 2. Tool matching requires measured performance against a defined comparison criterion, not simply a shared recipe or uptime figure.

06 / Answers and explanations / continued

2. matching criterion

Turn 4. The matching criterion allows no more than two nanometers between means; the observed difference is four.

3. actual value

Turn 6. Actual value concerns the measured outcome; passing the mean specification does not remove the separate difference between tools.

4. recipe revision

Turn 8. The recipe revision identifies the shared controlled input, but cannot stand in for unsupplied individual wafer evidence.

5. chamber condition

Turn 10. Chamber condition is a possible contributor to tool behavior that must be assessed rather than assumed from the recipe.

6. calibration status

Turn 12. Calibration status concerns the documented measurement basis and is relevant to interpreting the observed tool difference.

7. availability

Turn 14. Availability is the available-time fraction on the stated basis, not a measure of matching, yield, or approval.

8. equipment state

Turn 16. Equipment state records the tool's operational condition; an available state does not establish qualified process performance.

9. process transfer

Turn 18. Process transfer moves a defined process to another tool and requires its own relevant assessment and approval.

10. release status

Turn 20. Release status records authorization for the intended use, which remains absent despite the tool's availability and passing means.

Return to the conversation, page 76

06 / Answers and explanations / continued

F / Accuracy and scope

F1. Which calculation matches the availability basis?

C. 900 divided by 1,000 equals 90%.

Availability is available time over the defined window. Ten percent is the unavailable fraction, and utilization needs its own use-time data and basis.

Return to the exercise, page 79

F2. What remains necessary before calling B an approved interchangeable backup?

B. The required matching or qualification evidence and transfer approval

The case lacks matching acceptance and transfer approval, neither of which is replaced by uptime.

Return to the exercise, page 79

F / Guided speaking

1. specification

Engineer: Both means meet the 48-to-55 nm specification.

Turn 1. The specification applies to each sample mean, and both 50 and 53 fall within it.

2. matching criterion

Reviewer: But their three-nanometer gap misses our one-nanometer matching criterion.

Turn 2. The matching criterion applies to the three-nanometer difference, which exceeds the permitted one.

3. process transfer

Engineer: Then moving this process to the other tool remains a proposed process transfer.

Turn 3. Process transfer is the move between tools and requires its own assessment and authorization.

4. release status

Reviewer: Correct. Keep approval pending in the release status field.

Turn 4. Release status concerns authorization; passing individual means does not establish that approval.

Return to the conversation, page 80

06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Same recipe, different means". Victor: "For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent." Asha: [missing response] Victor: "We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions." Which response belongs in the gap?

D. Label it availability and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.

A Not here. This line belongs to turn 6, after Victor says: "Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Victor says: "I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Victor says: "We confirmed the same recipe revision and an agreed measurement basis. A's sample mean was ninety-eight nanometers, while B's was one hundred two." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Asha's response in turn 14. It connects Victor's preceding point with Victor's follow-up. Availability is the available-time fraction on the stated basis, not a measure of matching, yield, or approval.

Return to the exercise, page 80

07 / Answers and explanations

A / Read the situation

A1. Which test has passed?

A. Temperature cycling

The briefing identifies temperature cycling as the only completed passing test.

Return to the exercise, page 84

A2. What is the operating-life test status?

A. Running

The supplied plan still includes the test, and its stated status is running.

Return to the exercise, page 84

A3. Why is qualification complete unsupported?

D. Two required tests and final review remain open.

The defined plan requires evidence and final review that have not all been completed.

Return to the exercise, page 84

D / Response choices

D1. Which customer headline is supported?

D. Qualification in progress: temperature cycling passed; HTOL running; HAST not started.

The headline accurately separates each test's status without inventing a result or approval.

Return to the exercise, page 88

D2. What does the passing temperature-cycle result establish?

D. The stated test result under its protocol and scope

Temperature cycling, operating-life testing, and humidity-related stress address different test conditions. A shared name also does not establish equivalent package construction or scope.

Return to the exercise, page 88

07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Material placed around interconnections to support a package assembly. Which term fits?

C. underfill

A Not here. SiP: System in package, integrating multiple functions or components in one package. The usual combination is "evaluate a SiP design".

B Not here. HTOL: High-temperature operating life, an operating stress test under defined conditions. The usual combination is "report HTOL status".

C Correct. underfill: Material placed around interconnections to support a package assembly. The usual combination is "assess underfill integrity".

D Not here. failure analysis: Investigation of failed samples and their causes. The usual combination is "initiate failure analysis".

Return to the exercise, page 88

D4. Rebuild the exchange from "One completed test is not complete qualification". Diego: "The operating-life test is still running. The lab has not supplied its final result, so the customer should not read the current status as a pass." Mei: [missing response] Diego: "The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result." Which response belongs in the gap?

A. Label HTOL as running. If a progress checkpoint is available, identify it as interim information without treating it as the completed test outcome.

A Correct. This is Mei's response in turn 6. It connects Diego's preceding point with Diego's follow-up. HTOL is high-temperature operating life testing, which is currently running rather than completed in this case.

B Not here. This line belongs to turn 20, after Diego says: "I will change the headline to qualification in progress, list the three statuses, and keep the configuration and outstanding review visible. We will not announce release from this slide." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Diego says: "The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Diego says: "Which package construction should I name on the slide? Q7 alone may cover variants with different materials or assembly details." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 88

07 / Answers and explanations / continued

E / Conversation

1. qualification plan

Turn 2. The qualification plan defines all required tests and final review, so one passing test cannot complete it.

2. thermal cycling

Turn 4. Thermal cycling is the completed temperature-change stress test; its passing result remains valid within its defined protocol.

3. HTOL

Turn 6. HTOL is high-temperature operating life testing, which is currently running rather than completed in this case.

4. HAST

Turn 8. HAST is the highly accelerated stress test that has not started; absence of a result is not a pass or failure.

5. package construction

Turn 10. Package construction identifies the materials and structure covered by the evidence, preventing unsupported extension to other variants.

6. qualification vehicle

Turn 12. The qualification vehicle is the selected device or structure whose results support the defined qualification scope.

7. preconditioning

Turn 14. Preconditioning records show the defined preparation used before testing and form part of interpreting the evidence.

8. Qualification by similarity

Turn 16. Qualification by similarity requires an accepted technical rationale; no such substitution is established in this case.

9. Test coverage

Turn 18. Test coverage identifies which requirements or concerns the available tests address; final electrical testing does not replace every reliability stress.

07 / Answers and explanations / continued

10. qualification approval

Turn 20. Qualification approval is the authorized acceptance of the required evidence, which remains pending while tests and review are incomplete.

Return to the conversation, page 90

F / Accuracy and scope

F1. When could related evidence replace a required test in this case?

D. Only through the applicable accepted change or similarity process, which has not occurred

Related evidence requires an accepted relevance and authorization basis. Dimensions, document age, and a different passing test do not by themselves justify substitution.

Return to the exercise, page 92

F2. What should final review examine?

A. Required evidence, scope, configuration, open items, and approval conditions

Qualification approval depends on the required evidence and scope, not a favorable summary label.

Return to the exercise, page 92

F / Guided speaking

1. evidence

Engineer: Keep A's passing result as test evidence, not a claim that every requirement is complete.

Turn 1. The passing result is evidence for its defined test, not completion of the whole plan.

2. plan

Reviewer: B and C are still required by the approved plan.

Turn 2. The approved qualification plan still requires these tests; neither has an accepted replacement.

3. rationale

Engineer: We would need an accepted technical rationale before substituting related results.

Turn 3. A justified and accepted rationale is necessary for a relevant substitution; none exists here.

4. approval

Reviewer: Until the tests and final review are complete, do not announce qualification approval.

Turn 4. Qualification approval remains pending while required tests and final review are incomplete.

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07 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "One completed test is not complete qualification". Diego: "The assembly preparation records also belong in the package. I do not want the final review to discover that important preparation details were omitted from the summary." Mei: [missing response] Diego: "Sales wants to reuse results from an older package. I will refer that request for review; we have no accepted substitution for the open tests." Which response belongs in the gap?

B. Include the required preconditioning records where the plan calls for them. A passing result needs the relevant preparation and test context to be interpreted correctly.

A Not here. This line belongs to turn 12, after Diego says: "We should confirm the tested device or structure as well. Otherwise reviewers may not know how the selected samples relate to the configuration being discussed." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Mei's response in turn 14. It connects Diego's preceding point with Diego's follow-up. Preconditioning records show the defined preparation used before testing and form part of interpreting the evidence.

C Not here. This line belongs to turn 10, after Diego says: "Which package construction should I name on the slide? Q7 alone may cover variants with different materials or assembly details." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Diego says: "The customer slide says Q7 qualification is complete because temperature cycling passed. We still have two other tests in the plan, so I need a more accurate headline." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 93

08 / Answers and explanations

A / Read the situation

A1. Which earlier date is requested?

B. October 8 tape-out

Sofia asks to move tape-out from October 15 to October 8, not to deliver finished devices then.

Return to the exercise, page 95

A2. What remains unresolved in the design checks?

B. Four DRC findings and one LVS mismatch

The briefing explicitly lists four design-rule findings and one layout-versus-schematic mismatch.

Return to the exercise, page 95

A3. What is the capacity status?

D. A possible earlier slot, not reserved or accepted

A mentioned opportunity is not a recorded capacity commitment or approval.

Return to the exercise, page 95

D / Response choices

D1. Which reply to the earlier-date request is supported?

A. October 8 remains conditional on check closure, the required handoff, and capacity confirmation.

The reply preserves the separate technical and capacity prerequisites without inventing acceptance.

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D2. What does an LVS mismatch concern?

A. A difference between extracted layout connectivity and the schematic

LVS compares the layout-derived circuit with the schematic. Spacing rules, timing slack, and overlay belong to different verification or measurement questions.

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08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Process design kit: process-specific models, rules, and design resources. Which term fits?

A. PDK

A Correct. PDK: Process design kit: process-specific models, rules, and design resources. The usual combination is "confirm the PDK revision".

B Not here. engineering change order: A controlled record authorizing a defined engineering change, abbreviated ECO. The usual combination is "approve an engineering change order".

C Not here. engineering lot: A lot used for development or engineering evaluation. The usual combination is "plan an engineering lot".

D Not here. sign-off: The required formal completion and approval of defined checks. The usual combination is "complete design sign-off".

Return to the exercise, page 99

D4. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions." Ken: [missing response] Sofia: "Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled." Which response belongs in the gap?

D. Keep the PDK revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.

A Not here. This line belongs to turn 18, after Sofia says: "I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Sofia says: "Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Sofia says: "Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ken's response in turn 6. It connects Sofia's preceding point with Sofia's follow-up. The PDK supplies process-specific design resources; its agreed revision must remain connected to the verification results.

Return to the exercise, page 99

08 / Answers and explanations / continued

E / Conversation

1. tape-out

Turn 2. Tape-out is the design-data release milestone being requested, not a confirmed wafer start or finished-sample date.

2. sign-off

Turn 4. Sign-off is the required completion and approval of the defined checks; the complete package is not yet available here.

3. PDK

Turn 6. The PDK supplies process-specific design resources; its agreed revision must remain connected to the verification results.

4. DRC

Turn 8. DRC checks the layout against applicable design rules; its open findings need documented resolution rather than a changed headline.

5. LVS

Turn 10. LVS compares extracted layout connectivity with the schematic and therefore addresses a different verification question from geometric rule checks.

6. capacity reservation

Turn 12. Capacity reservation is an actual allocation; a possible earlier slot has not been reserved or accepted in this case.

7. Data acceptance

Turn 14. Data acceptance confirms the receiving process's requirements have been met; transmitting files alone does not establish it.

8. waiver

Turn 16. A waiver is an authorized exception to a specific requirement; none is approved for the open findings here.

9. wafer start

Turn 18. A wafer start is the lot's entry into fabrication and is distinct from the earlier design-data handoff milestone.

08 / Answers and explanations / continued

10. delivery commitment

Turn 20. A delivery commitment must identify an accepted deliverable and date; neither a requested tape-out nor a possible slot guarantees finished samples.

Return to the conversation, page 101

F / Accuracy and scope

F1. Why retain the PDK and rule-deck versions with results?

B. The check result depends on the process-specific definitions actually used.

Traceable versions identify the actual basis of the result. A filename is insufficient, and neither a later release nor an earlier clean run establishes unchanged applicability.

Return to the exercise, page 105

F2. Which statement about tape-out is accurate?

C. It is a design-data handoff milestone, not proof of wafer start or sample availability.

Fabrication, test, qualification, and delivery remain distinct activities after the design-data handoff.

Return to the exercise, page 105

F / Guided speaking

1. tape-out

Designer: Keep November third attached to the requested design-data handoff: tape-out.

Turn 1. Tape-out identifies the requested design-data handoff, not a later manufacturing or delivery event.

2. capacity reservation

Liaison: The possible slot is not yet an allocated capacity reservation.

Turn 2. Capacity reservation requires a confirmed allocation; a possible slot has not reached that status.

3. wafer start

Designer: I will not tell the customer that fabrication begins then; we have no confirmed wafer start.

Turn 3. A wafer start is distinct from the requested design-data handoff and is not confirmed here.

08 / Answers and explanations / continued

4. delivery commitment

Liaison: And discussing a slot does not establish a finished-sample delivery commitment.

Turn 4. A delivery commitment needs an accepted deliverable and date, neither of which follows from discussing a slot.

Return to the conversation, page 106

F / Rebuild the exchange

1. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "Who confirms that the foundry has accepted the data package? I do not want a file-transfer receipt reported as acceptance of the design handoff." Ken: [missing response] Sofia: "Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it." Which response belongs in the gap?

C. Exactly. Data acceptance is a separate handoff status. Record what the foundry needs and who confirms it, rather than assume a sent dataset has cleared the receiving review.

A Not here. This line belongs to turn 8, after Sofia says: "Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Sofia says: "We still have four design-rule findings and one layout-versus-schematic mismatch. I want to know whether those can remain open while we secure the opportunity." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Ken's response in turn 14. It connects Sofia's preceding point with Sofia's follow-up. Data acceptance confirms the receiving process's requirements have been met; transmitting files alone does not establish it.

D Not here. This line belongs to turn 2, after Sofia says: "Can we move tape-out from October fifteenth to October eighth? The foundry mentioned an earlier slot, and the customer is asking whether that means samples can arrive sooner." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 106

Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A semiconductor fabrication facility. Which term fits?

A. fab

A Correct. fab: A semiconductor fabrication facility. The usual combination is "move material through the fab".

B Not here. process module: A related group of operations within the fabrication flow. The usual combination is "qualify a process module".

C Not here. MOL: Middle of line, connecting devices to the interconnect structure. The usual combination is "review MOL interfaces".

D Not here. lot hold: A restriction preventing a lot from proceeding without disposition. The usual combination is "place a lot on hold".

Return to the exercise, page 28

2. A colleague needs the term for this meaning: The controlled settings and analysis used to obtain a measurement. Which term fits?

D. measurement recipe

A Not here. lithography: Pattern formation using exposure of a sensitive layer. The usual combination is "review lithography performance".

B Not here. optical proximity correction: Layout adjustments compensating for imaging effects, abbreviated OPC. The usual combination is "review optical proximity correction".

C Not here. overlay: Alignment accuracy between patterned layers. The usual combination is "measure overlay".

D Correct. measurement recipe: The controlled settings and analysis used to obtain a measurement. The usual combination is "confirm the measurement recipe".

Return to the exercise, page 28

Review 1 / Explanations / continued

3. Rebuild the exchange from "Back-end does not always mean packaging". Eric: "I also see that inspection review is pending. The lot cannot simply move to its next listed operation because someone has already processed the current step." Min: [missing response] Eric: "Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay." Which response belongs in the gap?

A. It is on a lot hold. That restriction remains in place until the required review and authorized decision. Processed, inspected, and released are different status claims.

A Correct. This is Min's response in turn 10. It connects Eric's preceding point with Eric's follow-up. A lot hold restricts further movement until the required review and authorized disposition occur.

B Not here. This line belongs to turn 4, after Eric says: "That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Eric says: "The revised message will locate L24 in wafer fabrication at M2, explain the inspection hold, and identify the next review. It will leave shipment timing uncommitted." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Eric says: "What later stages should I distinguish in the message? I need to explain why an M2 status does not yet give the customer a finished-product date." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 28

Review 1 / Explanations / continued

4. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "I will include the reticle revision in the context record as well. That gives the investigation the configuration information without implying the reticle has been found defective." Theo: [missing response] Lina: "The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open." Which response belongs in the gap?

A. Good. The reticle identity belongs with the layer history. Configuration details help evaluate hypotheses, but listing a component does not make it the cause.

A Correct. This is Theo's response in turn 18. It connects Lina's preceding point with Lina's follow-up. The reticle carries the lithographic pattern; identifying its revision provides context without proving it caused the observation.

B Not here. This line belongs to turn 12, after Lina says: "The review also asks whether this is an alignment problem. Can we clarify that separately? I do not want the CD figure copied into an overlay conclusion." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Lina says: "The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Lina says: "That locates the result, but I still need to know whether the two measurements were obtained on a comparable basis before calling it a physical process shift." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 28

Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Chemical vapor deposition, forming a film through chemical reactions of vapor precursors. Which term fits?

D. CVD

A Not here. overetch: Additional etching beyond a defined clearing or endpoint condition. The usual combination is "assess overetch effects".

B Not here. nominal setting: The chosen central or reference process setting. The usual combination is "test the nominal setting".

C Not here. step coverage: Film thickness on a specified feature relative to thickness on a reference surface. The usual combination is "evaluate step coverage".

D Correct. CVD: Chemical vapor deposition, forming a film through chemical reactions of vapor precursors. The usual combination is "evaluate CVD film properties".

Return to the exercise, page 56

2. A colleague needs the term for this meaning: A lot used as a stated comparison basis. Which term fits?

D. reference lot

A Not here. metrology: The science and practice of measurement. The usual combination is "review metrology consistency".

B Not here. SPC: Statistical process control, using data to monitor process behavior. The usual combination is "apply SPC".

C Not here. control signal: A result meeting a defined rule for investigation on a control chart. The usual combination is "respond to a control signal".

D Correct. reference lot: A lot used as a stated comparison basis. The usual combination is "identify the reference lot".

Return to the exercise, page 56

Review 2 / Explanations / continued

3. Rebuild the exchange from "A three-nanometer shift without a common basis". Lina: "Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement." Theo: [missing response] Lina: "The review also asks whether this is an alignment problem. Can we clarify that separately? I do not want the CD figure copied into an overlay conclusion." Which response belongs in the gap?

B. Exactly. Etch bias is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

A Not here. This line belongs to turn 14, after Lina says: "Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Theo's response in turn 10. It connects Lina's preceding point with Lina's follow-up. Etch bias concerns the dimensional change associated with pattern transfer and is one reason post-etch CD does not isolate exposure.

C Not here. This line belongs to turn 16, after Lina says: "If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 6, after Lina says: "Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting." It addresses that point, not the missing turn in the supplied exchange.

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Review 2 / Explanations / continued

4. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change." Ravi: [missing response] Elena: "The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis." Which response belongs in the gap?

D. Check the measurement system and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.

A Not here. This line belongs to turn 6, after Elena says: "The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Elena says: "Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Elena says: "The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ravi's response in turn 18. It connects Elena's preceding point with Elena's follow-up. The measurement system includes methods and conditions that affect whether the collected results can be compared consistently.

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Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Defects per unit area on a stated counting and inspection basis. Which term fits?

B. defect density

A Not here. post-process scan: Inspection performed after a specified process step. The usual combination is "compare post-process scans".

B Correct. defect density: Defects per unit area on a stated counting and inspection basis. The usual combination is "report defect density".

C Not here. defect classification: Assignment of detected features to defined categories. The usual combination is "refine defect classification".

D Not here. inspection sensitivity: The ability of an inspection method to detect specified features. The usual combination is "confirm inspection sensitivity".

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2. A colleague needs the term for this meaning: The share of a stated time basis during which equipment is used. Which term fits?

C. utilization

A Not here. tool matching: Demonstrating comparable tool performance against defined criteria. The usual combination is "assess tool matching".

B Not here. monitor wafer: A wafer used to track selected process or tool behavior. The usual combination is "review monitor-wafer results".

C Correct. utilization: The share of a stated time basis during which equipment is used. The usual combination is "state the utilization denominator".

D Not here. MTBF: Mean time between failures under a stated operating and event definition. The usual combination is "report MTBF".

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Review 3 / Explanations / continued

3. Rebuild the exchange from "Two percentage points, not a proven process gain". Elena: "Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference." Ravi: [missing response] Elena: "The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries." Which response belongs in the gap?

D. Use stratification to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

A Not here. This line belongs to turn 18, after Elena says: "We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 6, after Elena says: "The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Elena says: "The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ravi's response in turn 10. It connects Elena's preceding point with Elena's follow-up. Stratification separates the data into comparable product groups to examine the mix effect more directly.

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Review 3 / Explanations / continued

4. Rebuild the exchange from "Same recipe, different means". Victor: "The backup proposal remains useful because capacity is constrained, but it needs the appropriate technical assessment and approval before production relies on it." Asha: [missing response] Victor: "I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable." Which response belongs in the gap?

C. Treat it as a process transfer decision. The defined scope, matching evidence, and required authorization must be clear instead of being inferred from the existence of an alternative tool.

A Not here. This line belongs to turn 2, after Victor says: "Tool B has the same R7 recipe as A and was available for ninety percent of the window. The planning note calls it an interchangeable backup." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Victor says: "We confirmed the same recipe revision and an agreed measurement basis. A's sample mean was ninety-eight nanometers, while B's was one hundred two." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Asha's response in turn 18. It connects Victor's preceding point with Victor's follow-up. Process transfer moves a defined process to another tool and requires its own relevant assessment and approval.

D Not here. This line belongs to turn 8, after Victor says: "I will say both sample means meet that limit. We have not reviewed every individual measurement, so I cannot extend that claim to every wafer." It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Connecting a die face-down through conductive bumps or pillars. Which term fits?

C. flip-chip attachment

A Not here. QFN: Quad flat no-lead, a package format with exposed connection lands. The usual combination is "review the QFN package".

B Not here. thermal cycling: Repeated temperature changes under a specified test profile. The usual combination is "complete thermal cycling".

C Correct. flip-chip attachment: Connecting a die face-down through conductive bumps or pillars. The usual combination is "evaluate flip-chip attachment".

D Not here. test coverage: The requirements or failure concerns addressed by the performed tests. The usual combination is "assess qualification test coverage".

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2. A colleague needs the term for this meaning: The implemented rules used by a physical-verification tool. Which term fits?

D. rule deck

A Not here. mask data preparation: Processing layout data for mask manufacture. The usual combination is "schedule mask data preparation".

B Not here. wafer start: Entry of a wafer lot into the defined fabrication flow. The usual combination is "confirm the wafer start".

C Not here. foundry: A manufacturer producing semiconductor devices for other companies. The usual combination is "coordinate with the foundry".

D Correct. rule deck: The implemented rules used by a physical-verification tool. The usual combination is "identify the rule deck".

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Review 4 / Explanations / continued

3. Rebuild the exchange from "Same recipe, different means". Victor: "The observed difference needs investigation. Chamber history and maintenance condition may be relevant, but neither has been established as the cause from these numbers." Asha: [missing response] Victor: "I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?" Which response belongs in the gap?

D. Review chamber condition and the other appropriate records. Identical commanded settings do not guarantee identical physical conditions or measurement behavior.

A Not here. This line belongs to turn 6, after Victor says: "Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Victor says: "I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Victor says: "I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Asha's response in turn 10. It connects Victor's preceding point with Victor's follow-up. Chamber condition is a possible contributor to tool behavior that must be assessed rather than assumed from the recipe.

Return to the exercise, page 108

Review 4 / Explanations / continued

4. Rebuild the exchange from "A possible slot is not tape-out readiness". Sofia: "I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff." Ken: [missing response] Sofia: "The next update will separate check closure from slot confirmation. It will identify owners and show what remains conditional rather than announce the earlier date as secured." Which response belongs in the gap?

B. Correct. A wafer start is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.

A Not here. This line belongs to turn 6, after Sofia says: "The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Ken's response in turn 18. It connects Sofia's preceding point with Sofia's follow-up. A wafer start is the lot's entry into fabrication and is distinct from the earlier design-data handoff milestone.

C Not here. This line belongs to turn 8, after Sofia says: "Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Sofia says: "Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it." It addresses that point, not the missing turn in the supplied exchange.

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Additional case 1 / Explanations

Extended conversation

1. test bin

Turn 2. A test bin classifies the recorded test outcome; it does not by itself identify the physical cause of a failure.

2. site correlation

Turn 4. Site correlation examines whether the outcome follows a particular parallel test site, helping distinguish that pattern from the wafer's physical geography.

3. probe card

Turn 6. The probe card provides the electrical interface to wafer contacts; its condition and history can matter even when the test-program revision is unchanged.

4. contact resistance

Turn 8. Contact resistance concerns the electrical resistance at the probing connection; relevant measurements can help investigate a contact hypothesis without proving it from a label.

5. retest

Turn 10. Retest repeats testing under the applicable authorized conditions; it must preserve the first-pass record and identify the basis for interpreting a changed result.

6. denominator

Turn 14. The denominator identifies the population used in a yield calculation; first-pass and selected-retest populations must not be silently interchanged.

Return to the conversation, page 111

Transfer exchange

1. population

Yield engineer: Fifteen of twenty describes the selected retest population.

Turn 1. The retest population contains only selected initially failing die, not the whole wafer.

Additional case 1 / Explanations / continued

2. results

Test engineer: Preserve the original first-pass results.

Turn 2. Later testing must not erase the original observations or their reporting basis.

3. yield

Yield engineer: Do not label seventy-five percent as the whole-wafer yield.

Turn 3. Fifteen divided by twenty is the selected retest pass rate, not overall wafer yield.

4. plan

Test engineer: Interpret any changed result under the authorized retest plan.

Turn 4. The authorized plan determines the conditions and basis for using the retest evidence.

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Additional case 2 / Explanations

Extended conversation

1. setup slack

Turn 2. Setup slack compares the required and actual arrival timing for the setup check; negative slack means the stated constraint is not met in that scenario.

2. PVT corner

Turn 4. A PVT corner specifies process, voltage, and temperature assumptions for analysis; a pass at one corner does not automatically establish a pass at others.

3. timing exception

Turn 6. A timing exception changes how specified paths are analyzed and requires a justified functional basis; it is not a way to hide an inconvenient violation.

4. hold analysis

Turn 10. Hold analysis checks minimum-delay timing requirements and is distinct from setup analysis; a setup improvement does not automatically establish hold correctness.

5. parasitic extraction

Turn 12. Parasitic extraction provides modeled interconnect effects such as resistance and capacitance; its configuration must match the design being analyzed.

6. Timing closure

Turn 14. Timing closure is the process of meeting the defined timing requirements across the required analysis scope, not merely obtaining one favorable nominal report.

Return to the conversation, page 115

Transfer exchange

1. basis

Reviewer: Keep the violation visible until the exception has a justified basis.

Turn 1. An exception needs a valid functional rationale rather than merely a desire to remove the violation.

Additional case 2 / Explanations / continued

2. mode

Engineer: Check the path against the relevant operating mode.

Turn 2. Functional behavior in the applicable mode matters when deciding whether an exception is appropriate.

3. revision

Reviewer: Preserve the actual analysis inputs and configuration revision.

Turn 3. The timing result must remain linked to the precise configuration analyzed.

4. open

Engineer: Do not call timing closed while required checks remain open.

Turn 4. Incomplete required checks prevent a blanket claim that the defined timing scope is complete.

Return to the conversation, page 118

Additional case 3 / Explanations

Extended conversation

1. moisture sensitivity level

Turn 2. The moisture sensitivity level identifies a package's classification for moisture-related handling; it is not evidence that this reel's actual exposure stayed within its requirements.

2. floor life

Turn 4. Floor life concerns the permitted exposure under specified conditions before the relevant assembly process; a new bag seal alone does not establish a fresh exposure allowance.

3. cumulative exposure

Turn 6. Cumulative exposure concerns the relevant accumulated time and conditions after opening; an undocumented interval cannot be treated as zero simply because the reel was resealed.

4. humidity indicator card

Turn 8. A humidity indicator card provides an indication relevant to humidity within the package under its specified interpretation; it does not reconstruct the complete historical exposure record.

5. Shelf life

Turn 10. Shelf life concerns storage over a defined period under stated conditions; it is a different question from post-opening floor life before assembly.

6. disposition

Turn 14. Disposition is the authorized decision about the material's next route after reviewing the relevant evidence and requirements; resealing does not supply that decision.

Return to the conversation, page 120

Transfer exchange

1. history

Coordinator: Storage shelf life does not resolve the missing exposure history.

Turn 1. The storage period and post-opening handling record address different requirements.

Additional case 3 / Explanations / continued

2. requirements

Engineer: Identify the actual part, package, and applicable handling requirements.

Turn 2. The assessment must use the requirements for the specific material rather than a similar component.

3. record

Coordinator: Keep unknown time separate from a confirmed record.

Turn 3. An undocumented interval must not be treated as a verified value.

4. issue

Engineer: Await the authorized material decision before routine issue.

Turn 4. The unresolved handling status needs the applicable disposition before normal production issue.

Return to the conversation, page 123

Find a phrase by purpose

Wafer Fabrication Flow and Process Integration

Clarify the abbreviation: Here BEOL means back end of line.

Locate the work: The lot is still in wafer fabrication.

Name the layer: The current operation concerns metal layer M2.

Full phrase bank, page 7

Lithography, Reticles, and Critical Dimensions

Locate the dimension: Which layer and feature does the CD refer to?

Identify the stage: These measurements are from M2 after etch.

State the numbers: The reported means are 86 and 89 nm.

Full phrase bank, page 19

Deposition, Etch, CMP, and Process Windows

Identify the metric: The defined thickness-nonuniformity metric fell from 5% to 3%.

Keep the basis fixed: Both values use the same measurement definition.

State the condition: The result applies to the nominal setting only.

Full phrase bank, page 34

Metrology, SPC, and Yield Learning

Give the pilot counts: The pilot passed 92 of 100 tested die.

Give the reference counts: The reference passed 900 of 1,000 tested die.

Name the absolute change: The observed difference is two percentage points.

Full phrase bank, page 47

Find a phrase by purpose / continued

Defect Density and Cleanroom Contamination

Replace the vague label: The record shows particle-like surface detections, not a confirmed cleanroom source.

Give the count and area: There were 24 detections over 200 square centimeters inspected.

Name the density basis: That is 0.12 detections per square centimeter.

Full phrase bank, page 61

Equipment Uptime, Recipes, and Tool Matching

Confirm the recipe: Both tools used the identified R7 revision.

Separate inputs from outputs: Identical setpoints do not prove identical performance.

State the means: A measured 98 nm and B measured 102 nm.

Full phrase bank, page 73

Packaging, Test, and Reliability Qualification

Name the completed result: Q7 passed temperature cycling under the stated protocol.

State ongoing work: HTOL is still running.

State unstarted work: HAST has not started.

Full phrase bank, page 87

Foundry, Tape-Out, PDK, and Capacity Communication

Name the requested milestone: You are requesting October 8 tape-out.

Separate later dates: That is not a wafer-start or sample-delivery commitment.

State the open checks: Four DRC findings and one LVS mismatch remain open.

Full phrase bank, page 98

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References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] ASML. How Microchips Are Made.

Background on wafer processing, layers, lithography, and industry roles. The cases do not specify an actual fabrication flow or reproduce an equipment recipe. Source checked 30 September 2026.

https://www.asml.com/en/technology/all-about-microchips/how-microchips-are-made

[2] NIST/SEMATECH. Engineering Statistics Handbook, Section 6.3.1: What Are Control Charts?

Background on monitoring process behavior. The yield comparisons use invented data and do not claim a completed statistical process study. Source checked 30 September 2026.

https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc31.htm

[3] Texas Instruments. Reliability.

An industry example of reliability and qualification practice. The fictional three-test plan is not asserted as a universal qualification requirement. Source checked 30 September 2026.

https://www.ti.com/quality-reliability/reliability.html

[4] SkyWater SKY130 PDK Documentation. Process Design Rules.

An open example of process-specific design rules, layer definitions, and physical verification terminology. The tape-out case does not use or approve a real design. Source checked 30 September 2026.

https://skywater-pdk.readthedocs.io/en/main/rules.html

> **Companion course**
>
> The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.
>
> https://englishladder.com/efsp-semiconductor.html

Name the layer. Qualify the result. Align the handoff.

Eight semiconductor conversations connecting process context, measurement, yield, qualification, and delivery commitments.

Wafer Fabrication Flow and Process Integration

BEOL is still part of wafer fabrication in this conversation. It is not interchangeable with package assembly, although some organizations also use back-end for assembly and test. Expand the term and state the actual operation before communicating status.

Lithography, Reticles, and Critical Dimensions

The arithmetic difference is 3 nm. That does not prove a 3 nm physical process shift because the recipes and sampled sites differ. A post-etch CD combines the history of pattern formation and transfer; it does not identify exposure as the sole cause.

Deposition, Etch, CMP, and Process Windows

The metric improves by 2 percentage points, a 40% relative reduction from its initial 5% value. Neither calculation establishes the proposed window. The same measurement definition is used here; other nonuniformity formulas must not be mixed into the comparison.

Metrology, SPC, and Yield Learning

The pilot has 92% yield and the reference 90%. The difference is 2 percentage points, or about 2.22% relative to 90%. Different product mixes and sample sizes mean this observation alone does not establish a causal improvement or statistical significance.

Name the layer. Qualify the result. Align the handoff. / continued

Phil Stilwell

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